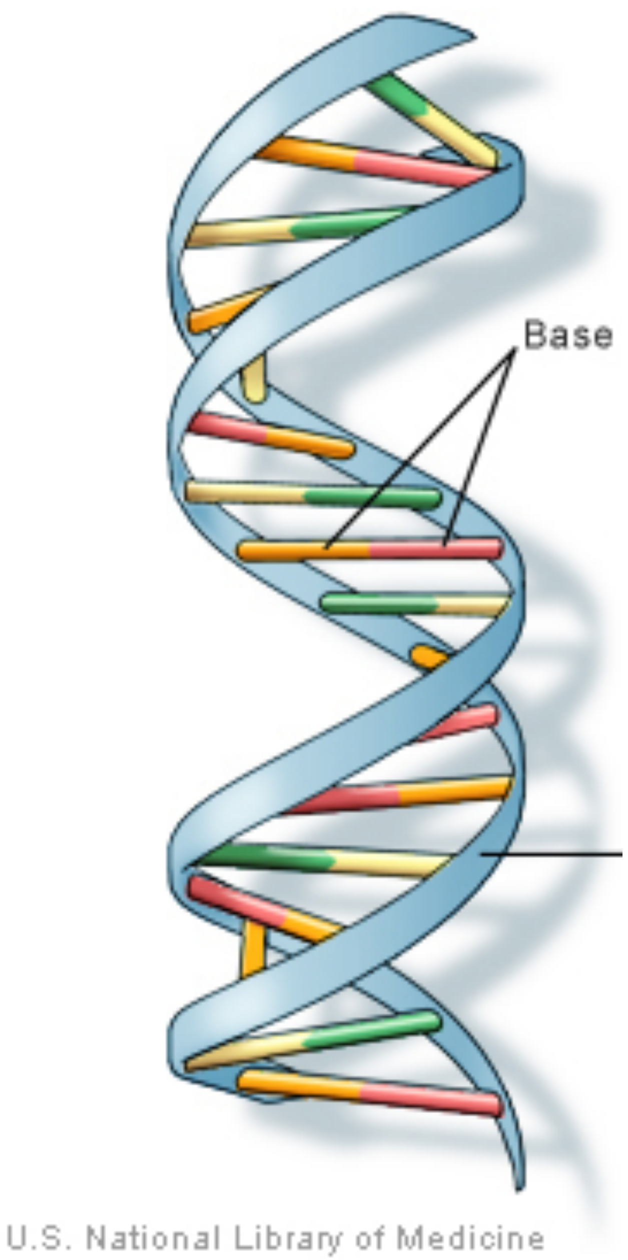
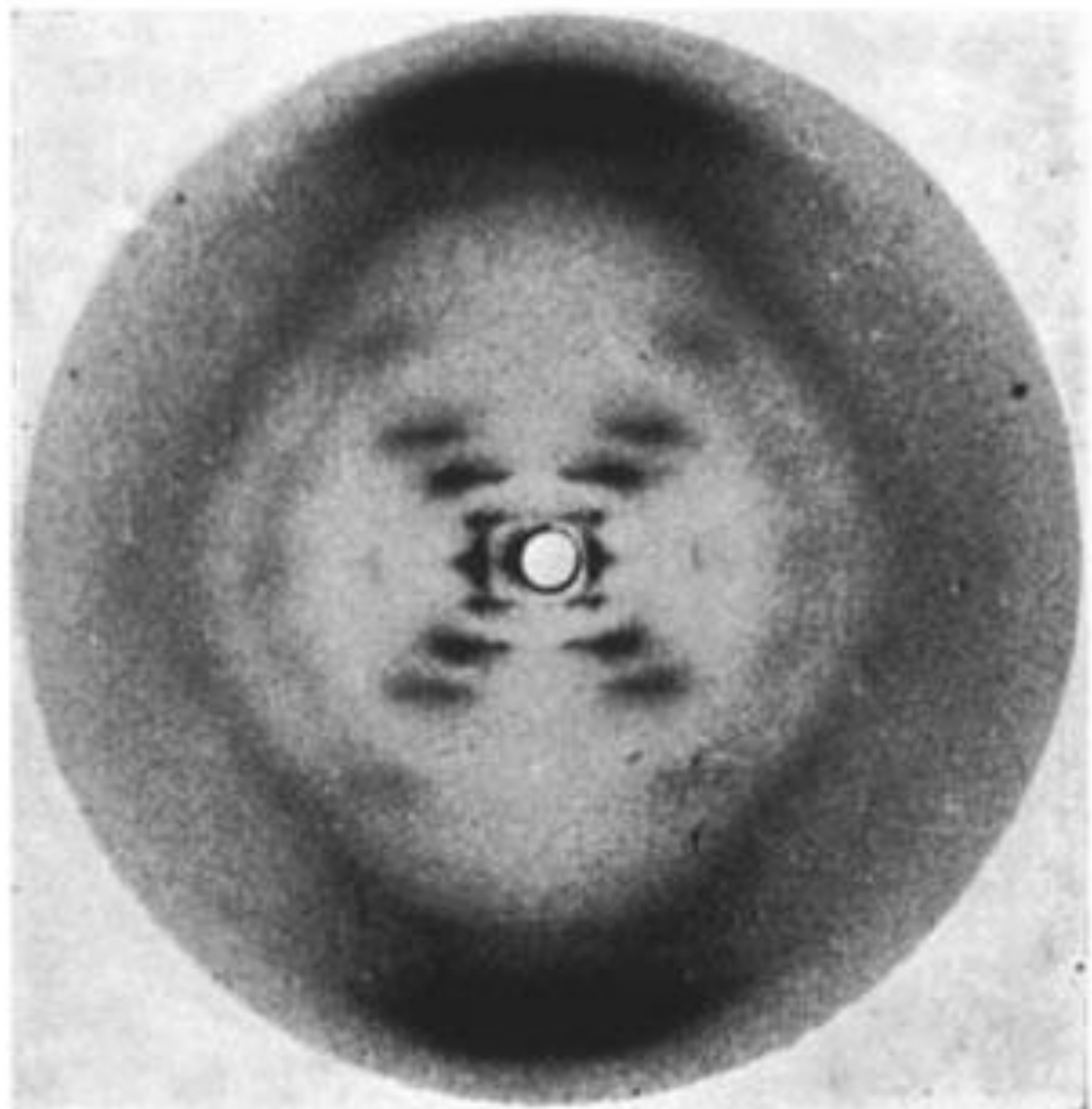
Inverse problems and imaging

- Recap of last class
- Inverse problems and DNA
- Another inverse problem in medical imaging. Light attenuation model and the Radon transform
- Working with images in Matlab

Photograph 51, FT, double helix

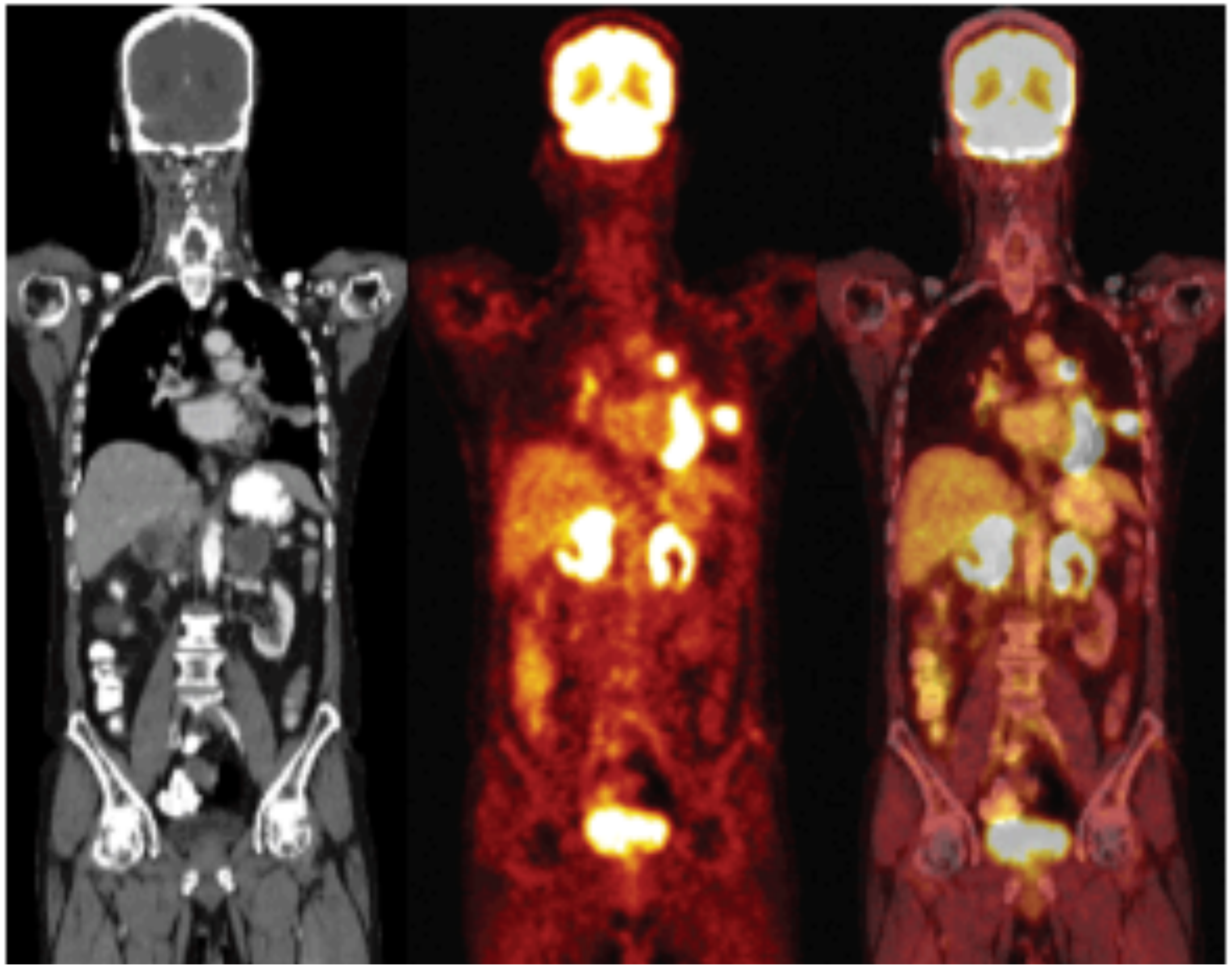




Medical imaging
“Reconstruction” problems

Medical imaging

- Many medical images result from solving an inverse problem



Enter patient: 65 year old man

- Persistent dry cough
- Mild fatigue
- Smoked a pack a day for a decade
- Lost 10 – 15 pounds in last six months

Patient's chest x-ray

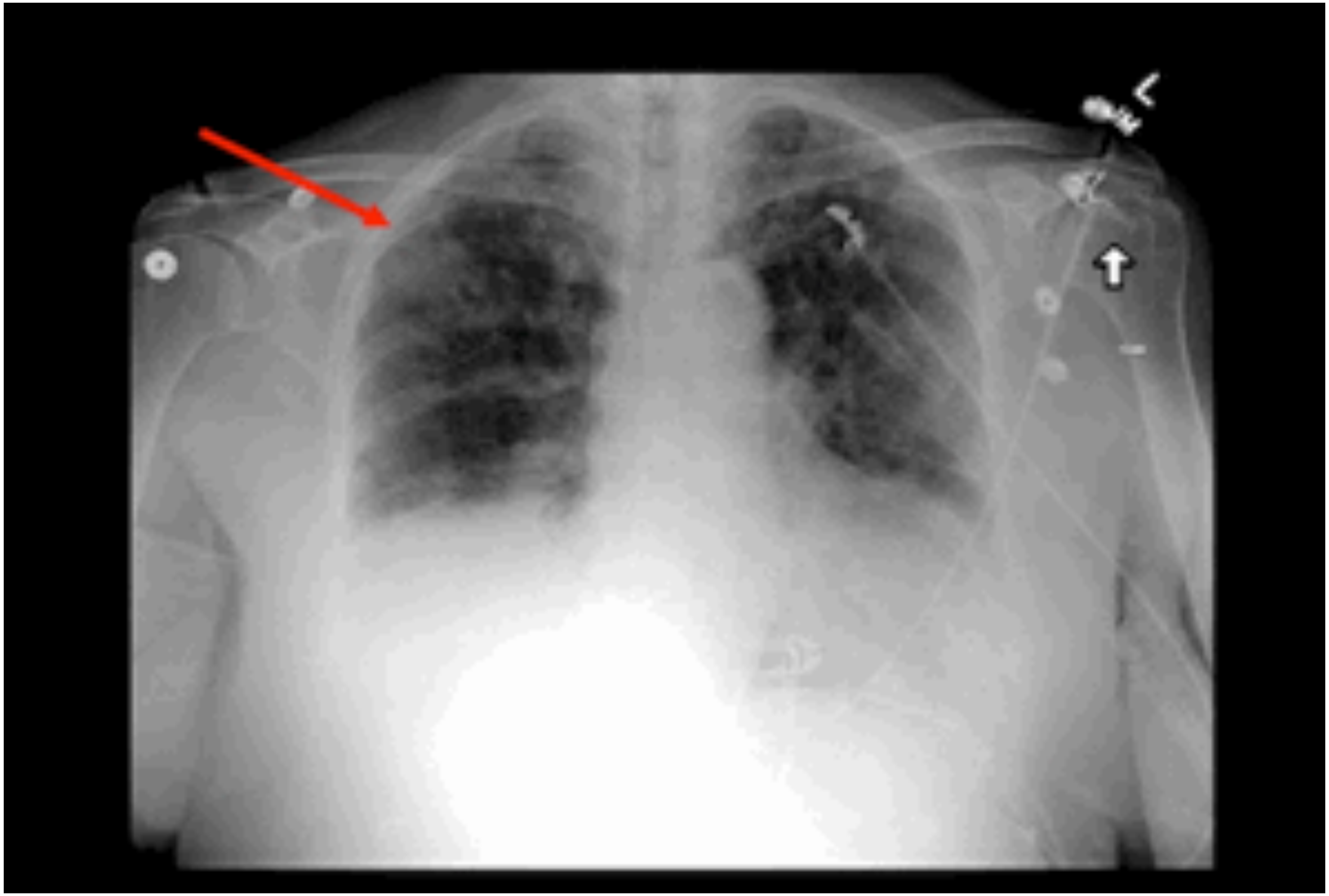
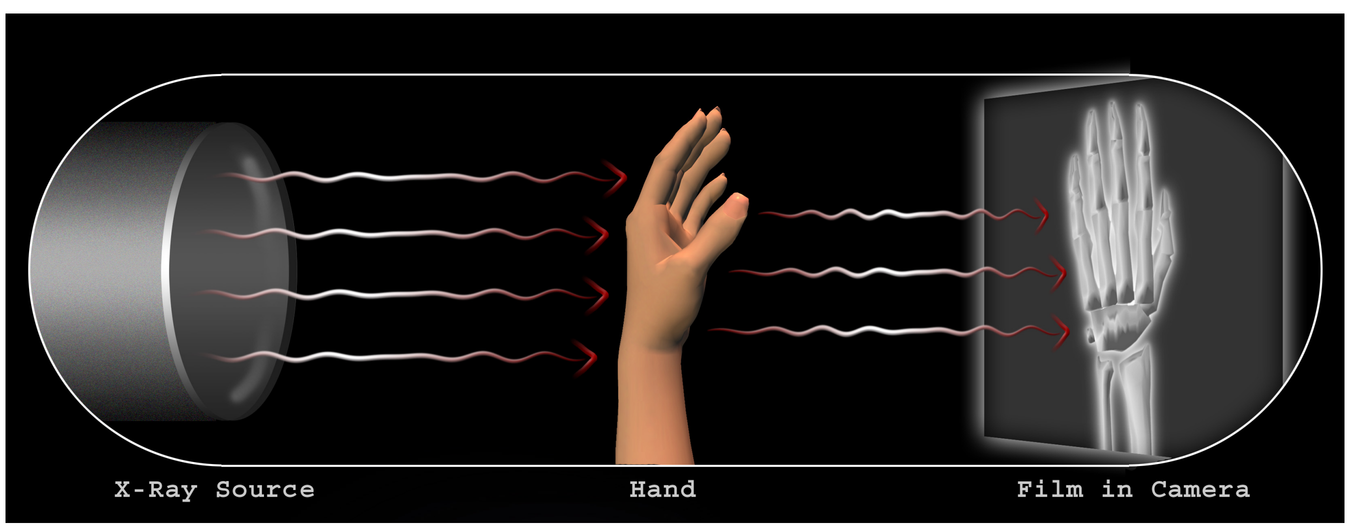


Photo credit: Dr. Joseph Levine

Conventional x-rays

Roentgen (1895):
experiments with electron beams
1901: first physics Nobel



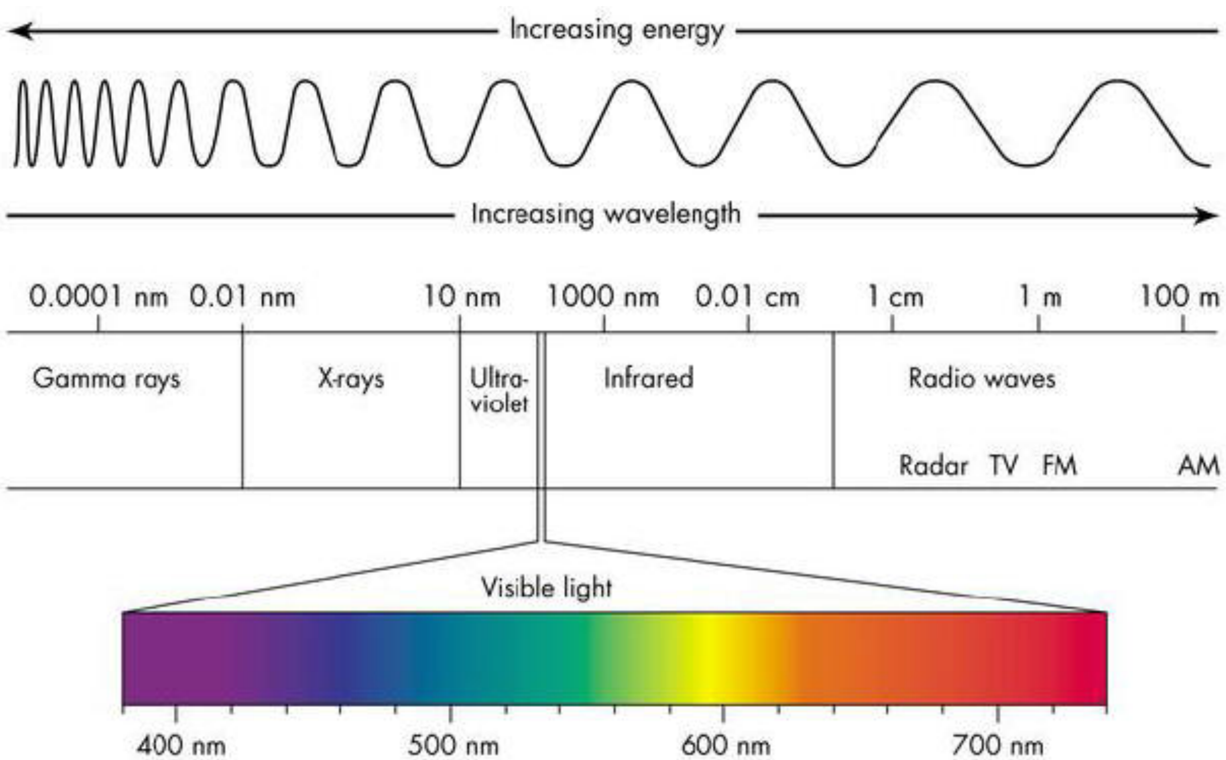
First radiographs

Roentgen’s wife’s hand
(1895)

Colleague’s hand
(1896)

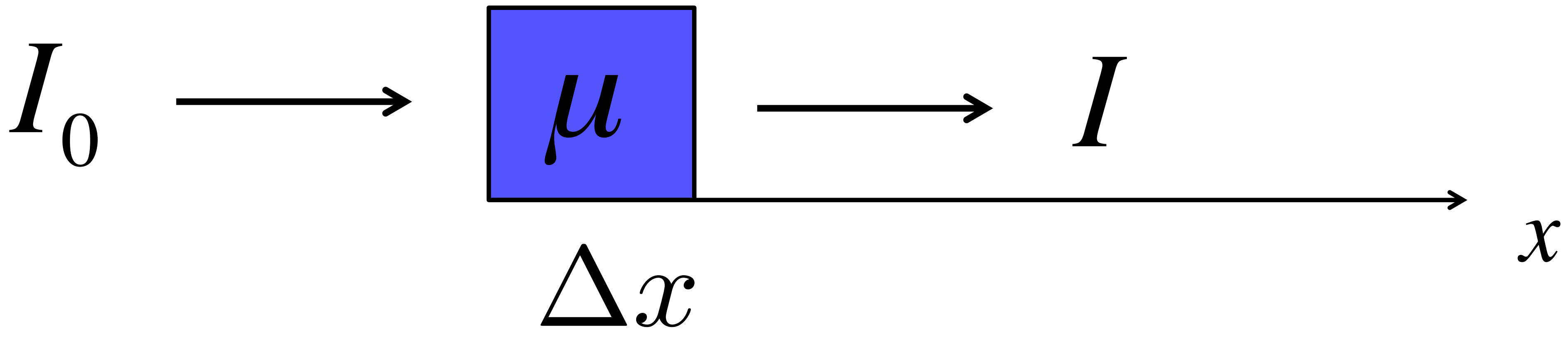
https://en.wikipedia.org/wiki/Wilhelm_Röntgen

Electromagnetic spectrum

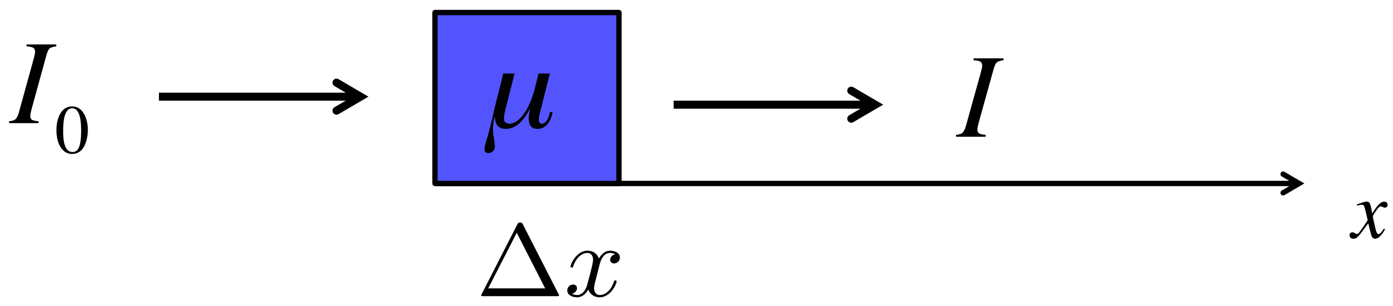


Material	Attenuation μ (1/cm)
Bone, cortical	6.9
Bone, trabecular	9.94
Brain	0.6
Breast	0.75
Cardiac	0.52
Connective tissue	1.57
Dentin	80
Enamel	120
Fat	0.48
Liver	0.5
Marrow	0.5
Muscle	1.09
Tendon	4.7
Soft tissue (average)	0.54
Water	0.0022

Activity: model for attenuation?



Attenuation: uniform material



$$\frac{dI}{dt} = -\mu I$$

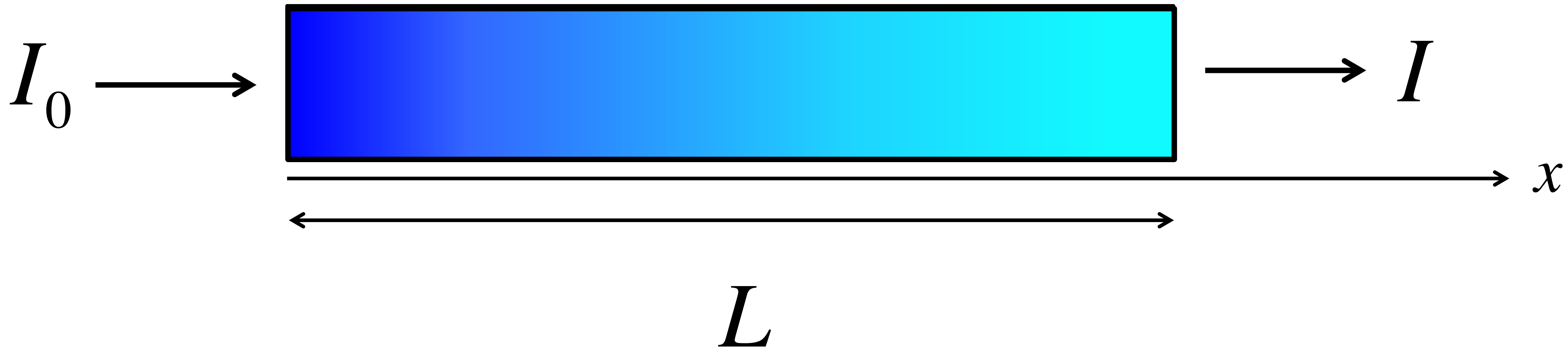
$$I(0) = I_0$$

$$I(x) = I_0 e^{-\mu x}$$

$$I_1 = I(\Delta x) = I_0 e^{-\mu \Delta x}$$

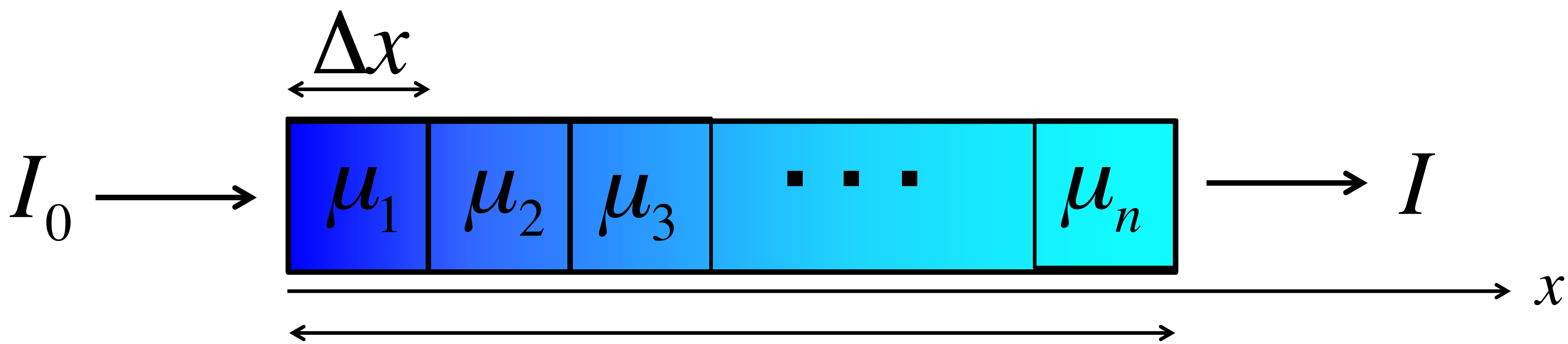
Attenuation: nonuniform material

$$\mu = \mu(x)$$



Attenuation: nonuniform material

$$\mu = \mu(x)$$



$$n\Delta x = L$$

Attenuation: nonuniform material

$$I_0 \longrightarrow \boxed{\mu_1} \longrightarrow I_1 \approx I_0 e^{-\mu_1 \Delta x}$$

$$I_1 \longrightarrow \boxed{\mu_2} \longrightarrow I_2 \approx I_1 e^{-\mu_2 \Delta x} \approx I_0 e^{-\mu_1 \Delta x} e^{-\mu_2 \Delta x}$$

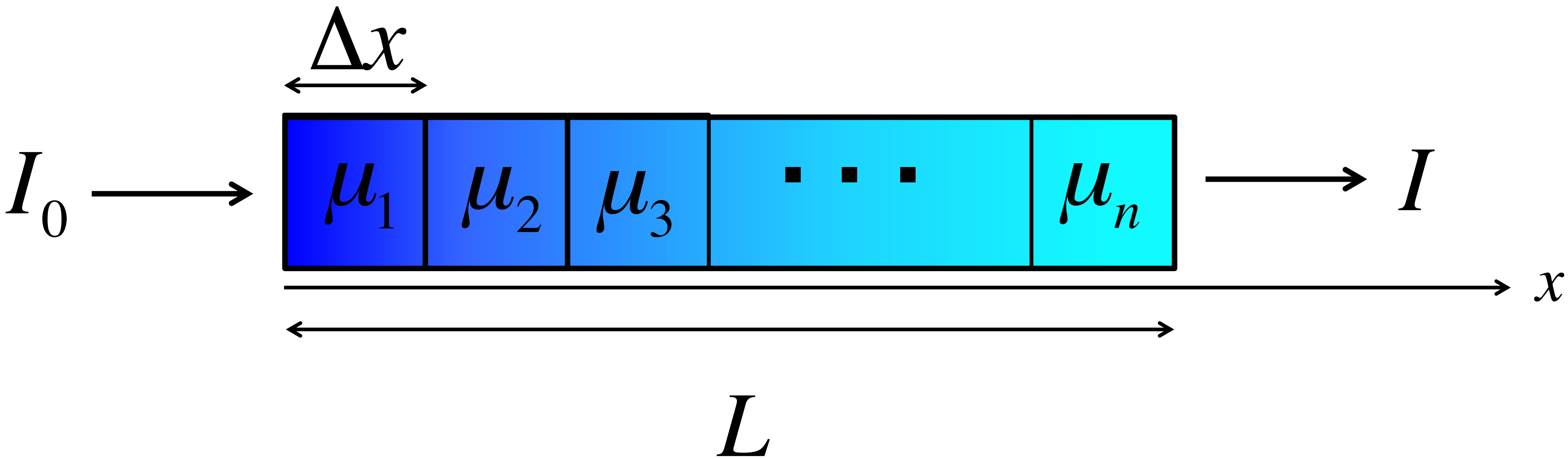
$$= I_0 e^{-(\mu_1+\mu_2)\Delta x}$$

$$\vdots$$

$$I_{n-1} \longrightarrow \boxed{\mu_n} \longrightarrow I_n \approx I_{n-1} e^{-\mu_n \Delta x} \approx I_0 e^{-\boxed{(\mu_1+\mu_2+\ldots+\mu_n)\Delta x}}$$

Attenuation: nonuniform material

$$\mu = \mu(x)$$

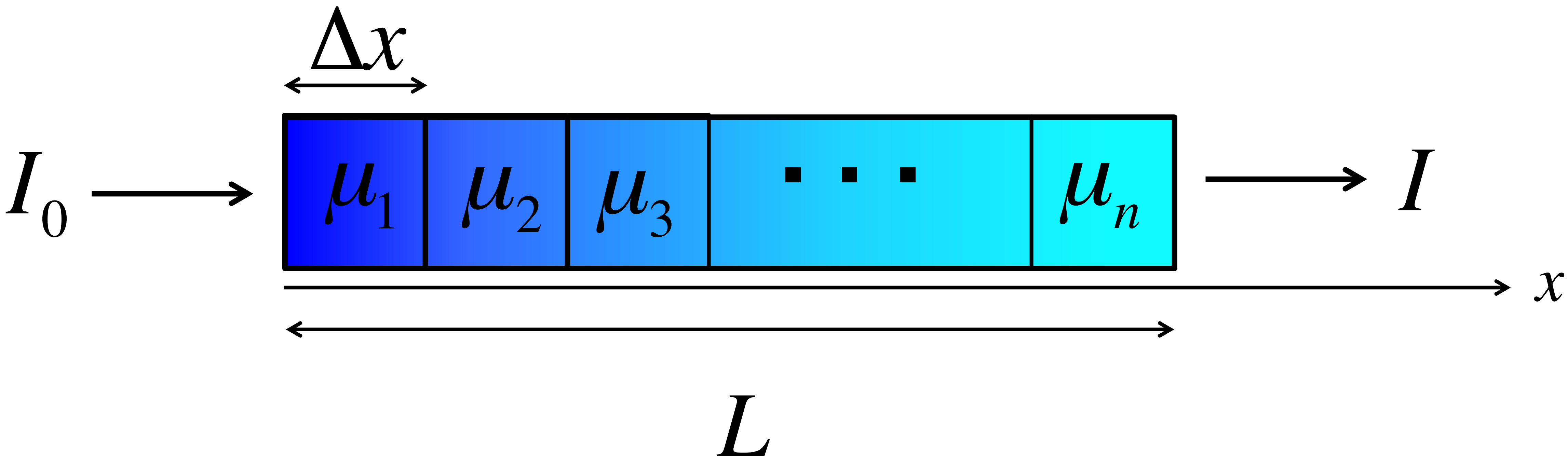


$$I \approx I_0 e^{-\boxed{(\mu_1 + \mu_2 + \ldots + \mu_n)\Delta x}}$$

Riemann Sum

Attenuation: nonuniform material

$$\mu = \mu(x)$$



$$\boxed{I = I_0 e^{-\int_0^L \mu(x)\, dx}}$$

What is known?

measureable output x-ray intensity

$$p = -\ln\left(\frac{I}{I_0}\right) = \int_0^L \mu(x)\,dx$$

known input x-ray intensity

x-ray projection

What is known? What is unknown?

$$\boxed{p} = \int_0^L \boxed{\mu(x)}\, dx$$

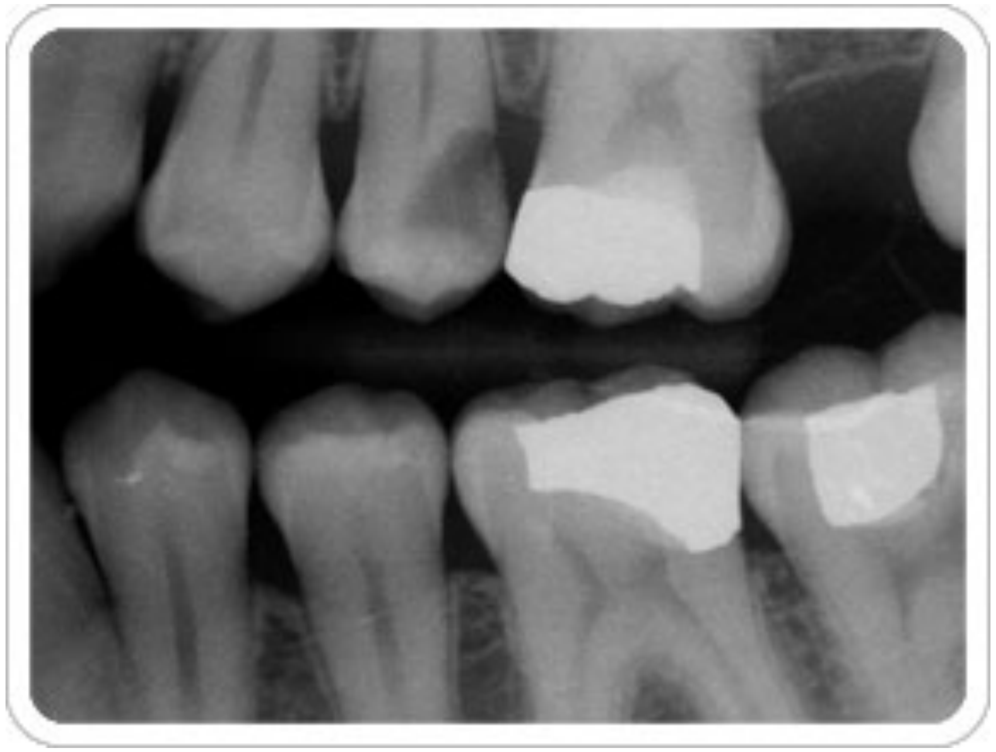
KNOWN!

UNKNOWN!

Inverse problem, image reconstruction problem

Given: p
Find: $\mu(x)$

X-rays: Shadows

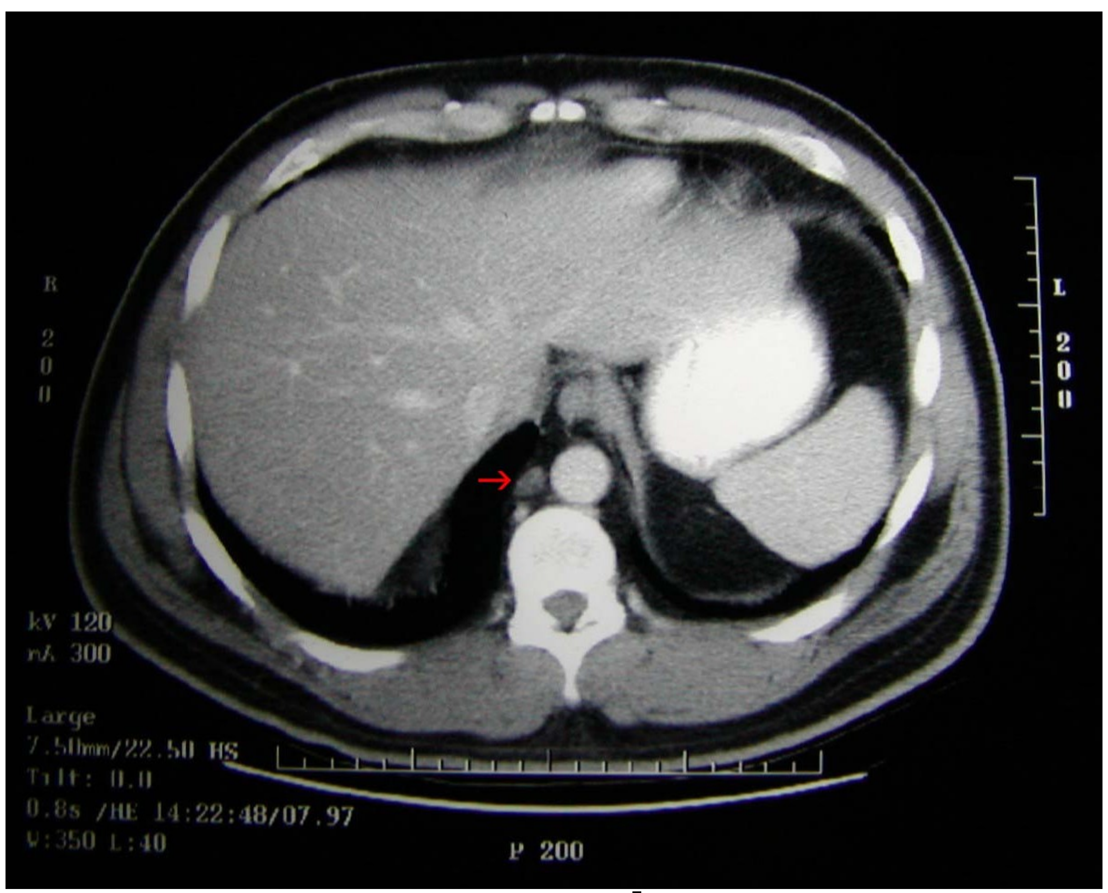


p *x-ray projection*

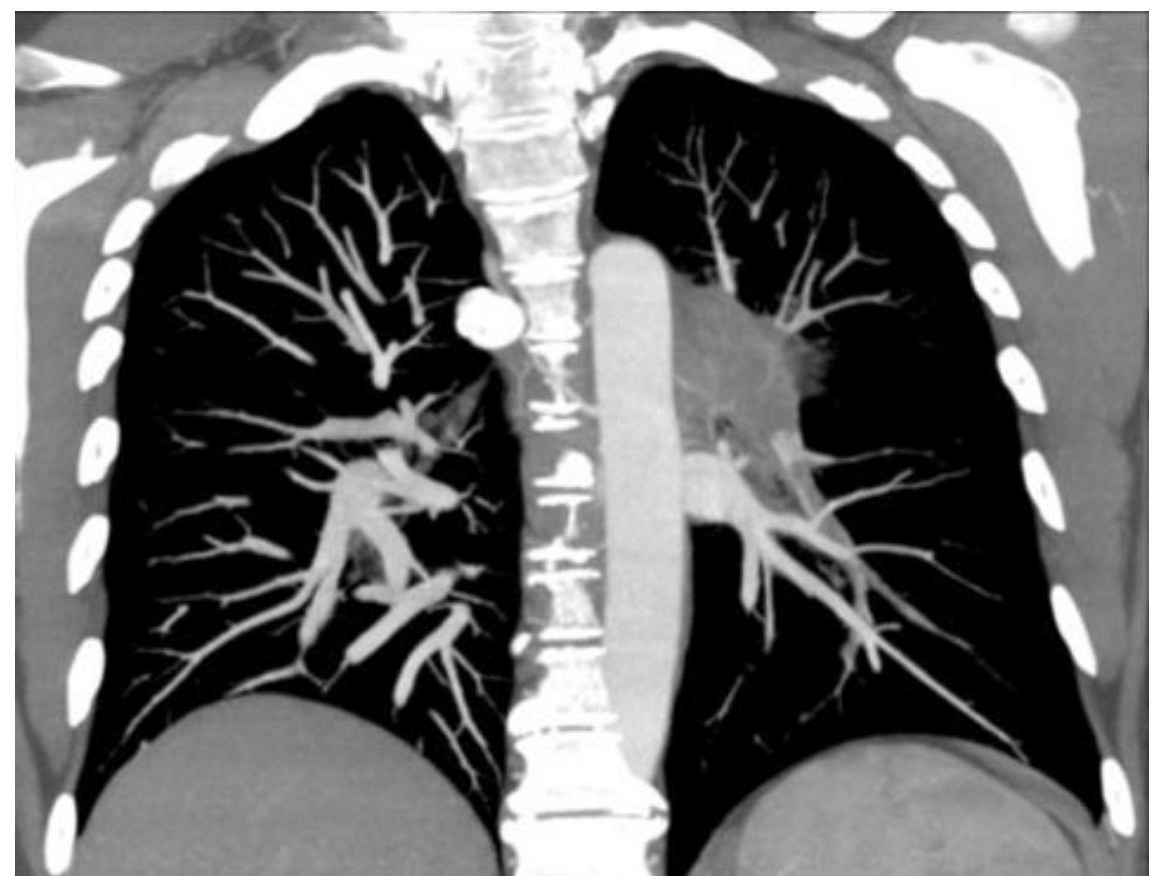
Tomography: solve for $\mu(x)$

Tomography: "Any of several techniques for making detailed x-rays of a predetermined plane section of a solid object while blurring out the images of other planes."

Chest Computed Tomography (CT) Images



1 cm carcinoma



pulmonary embolism

Patient’s CT Scan

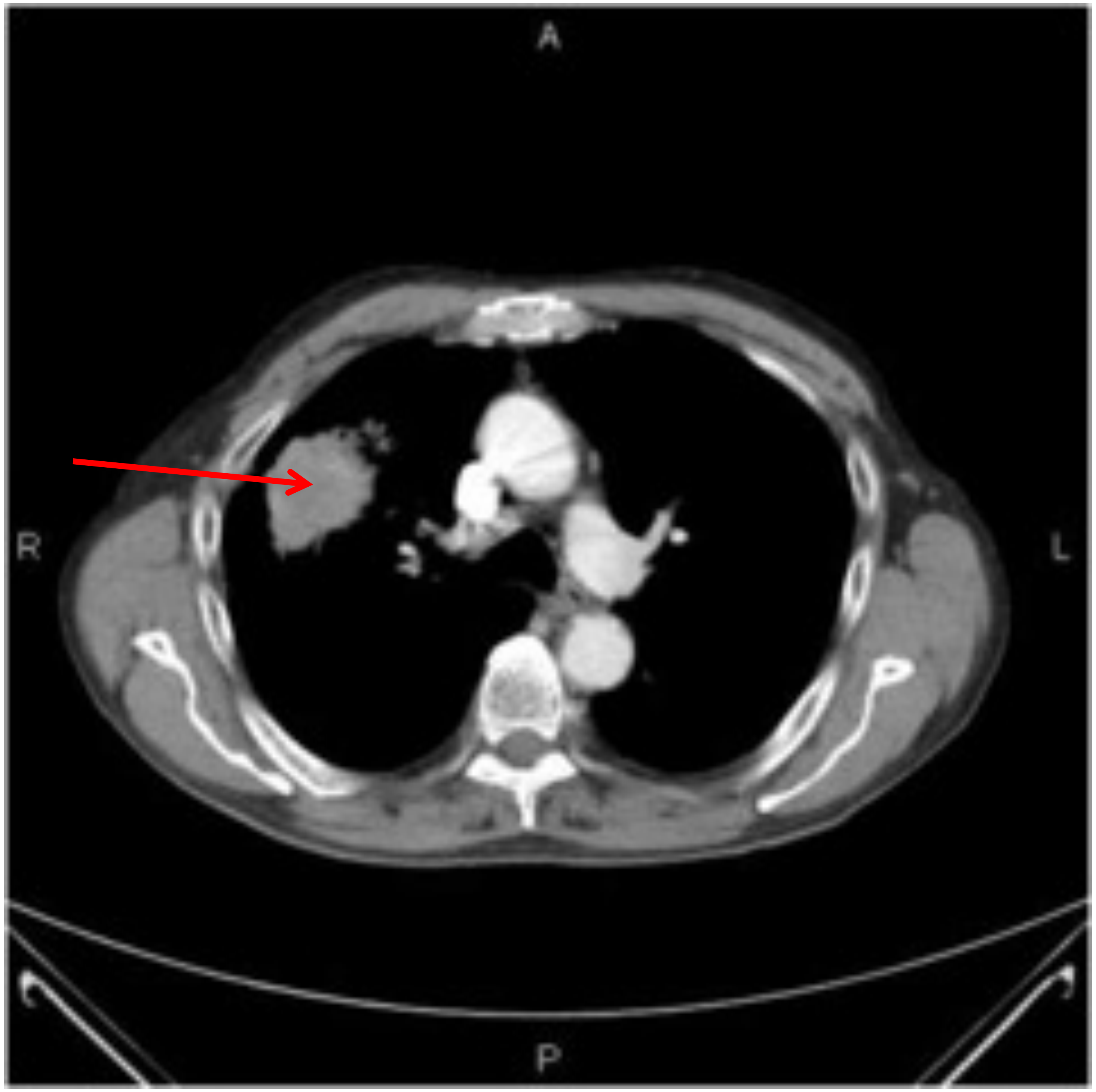
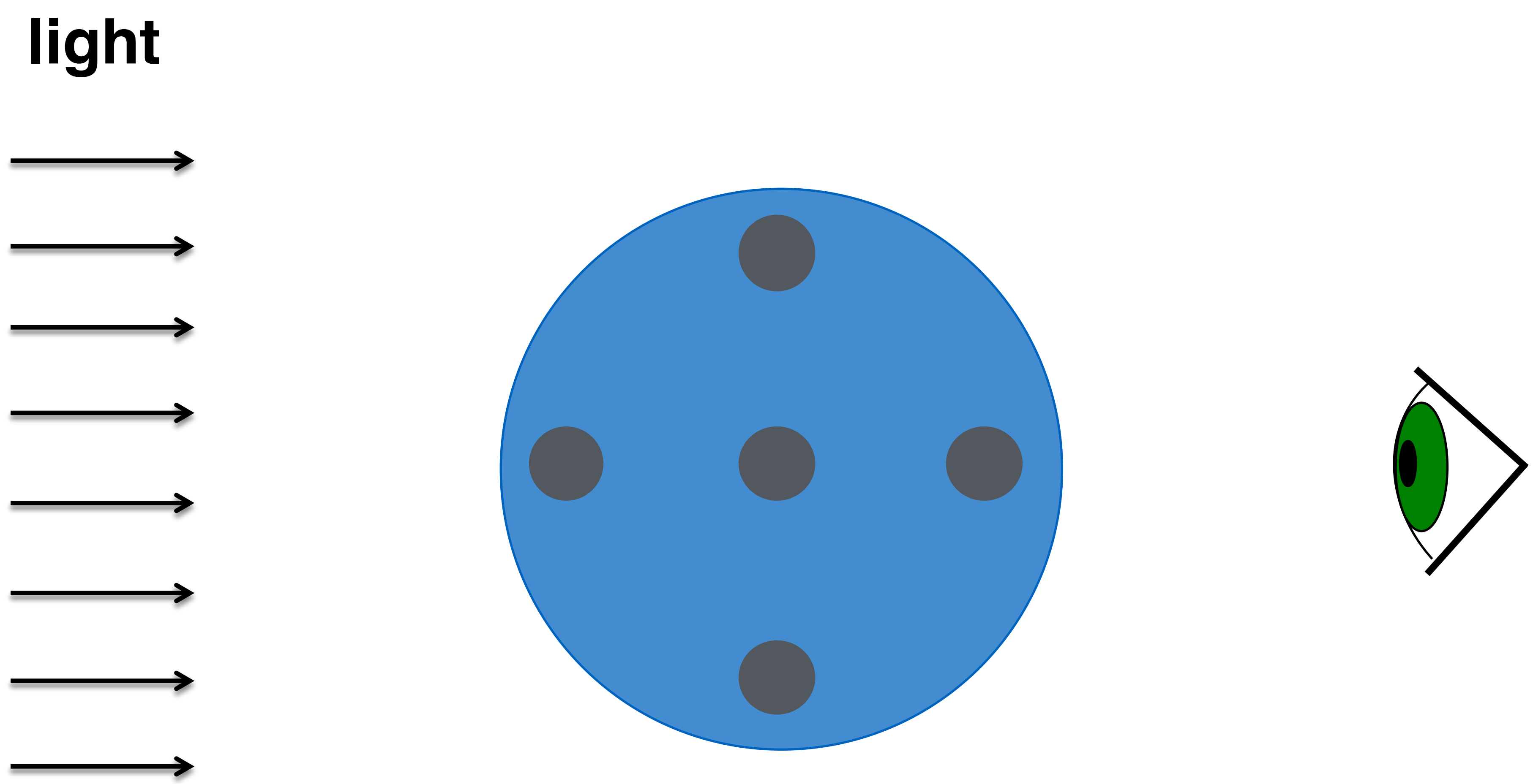


Photo credit: Dr. Joseph Levine

Basic idea: view 1

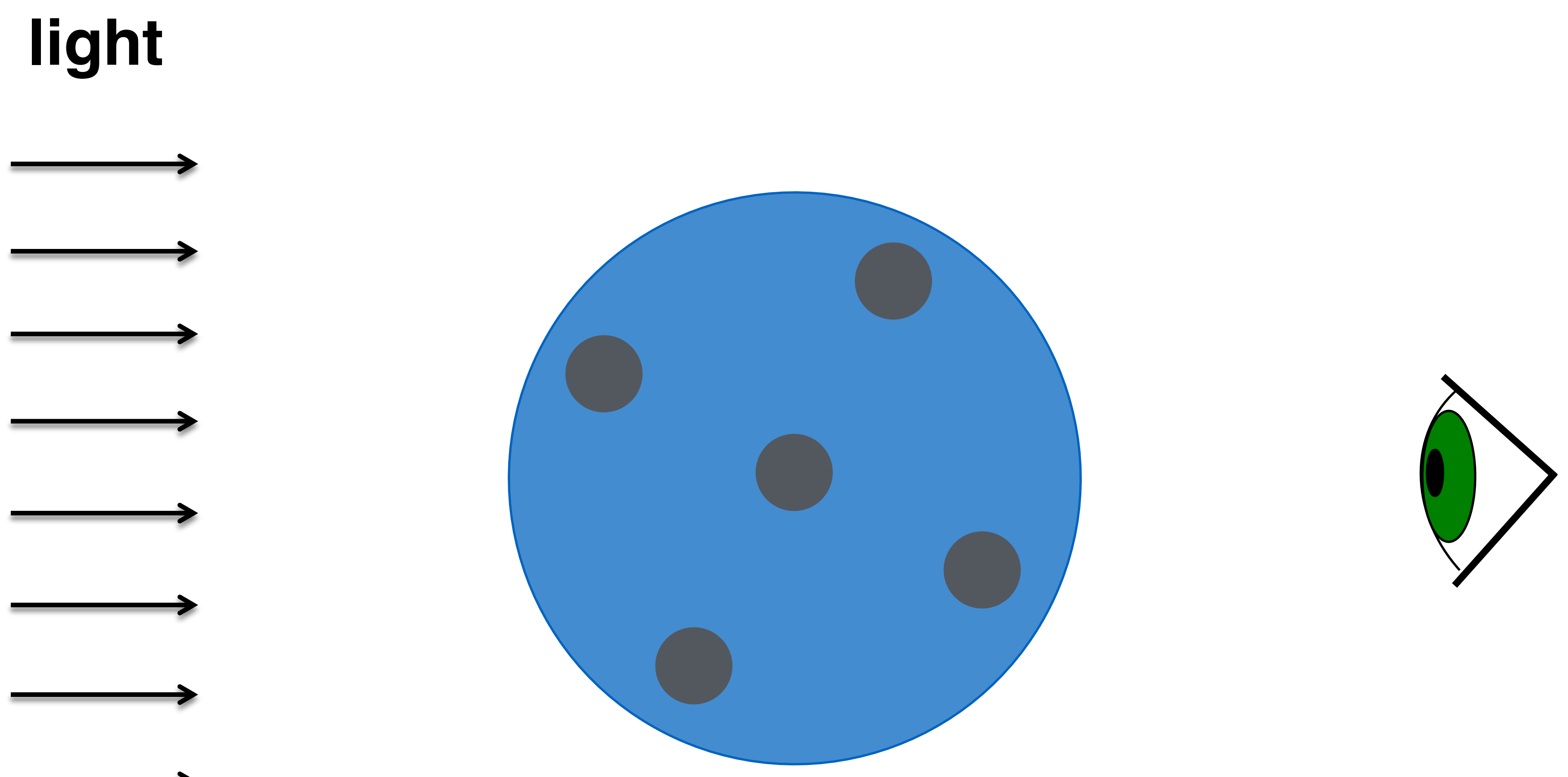
Semi-opaque cylinder with 5 smaller cylinders embedded



Top view

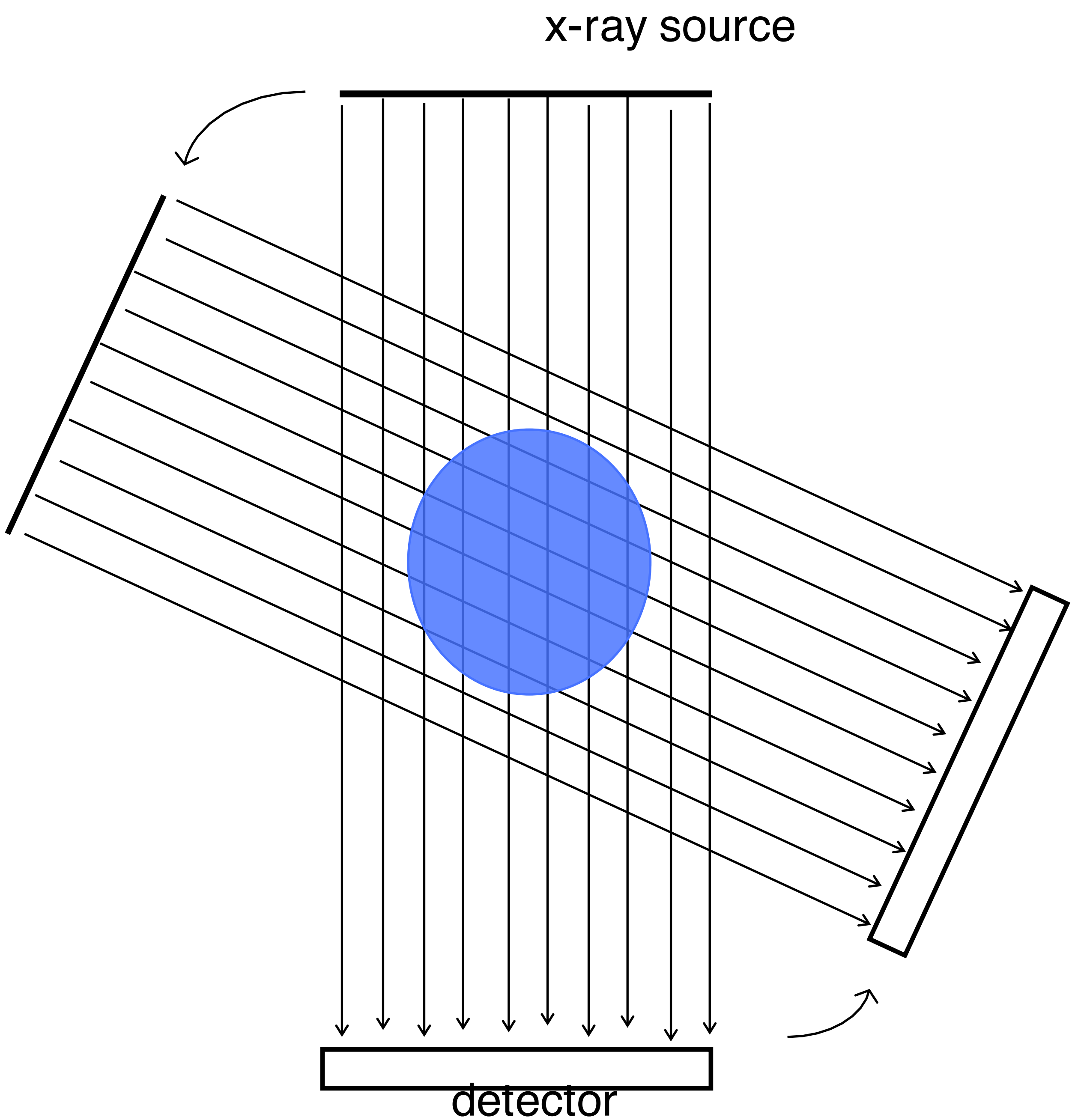
Basic idea: view 2

Semi-opaque cylinder with 5 smaller cylinders embedded

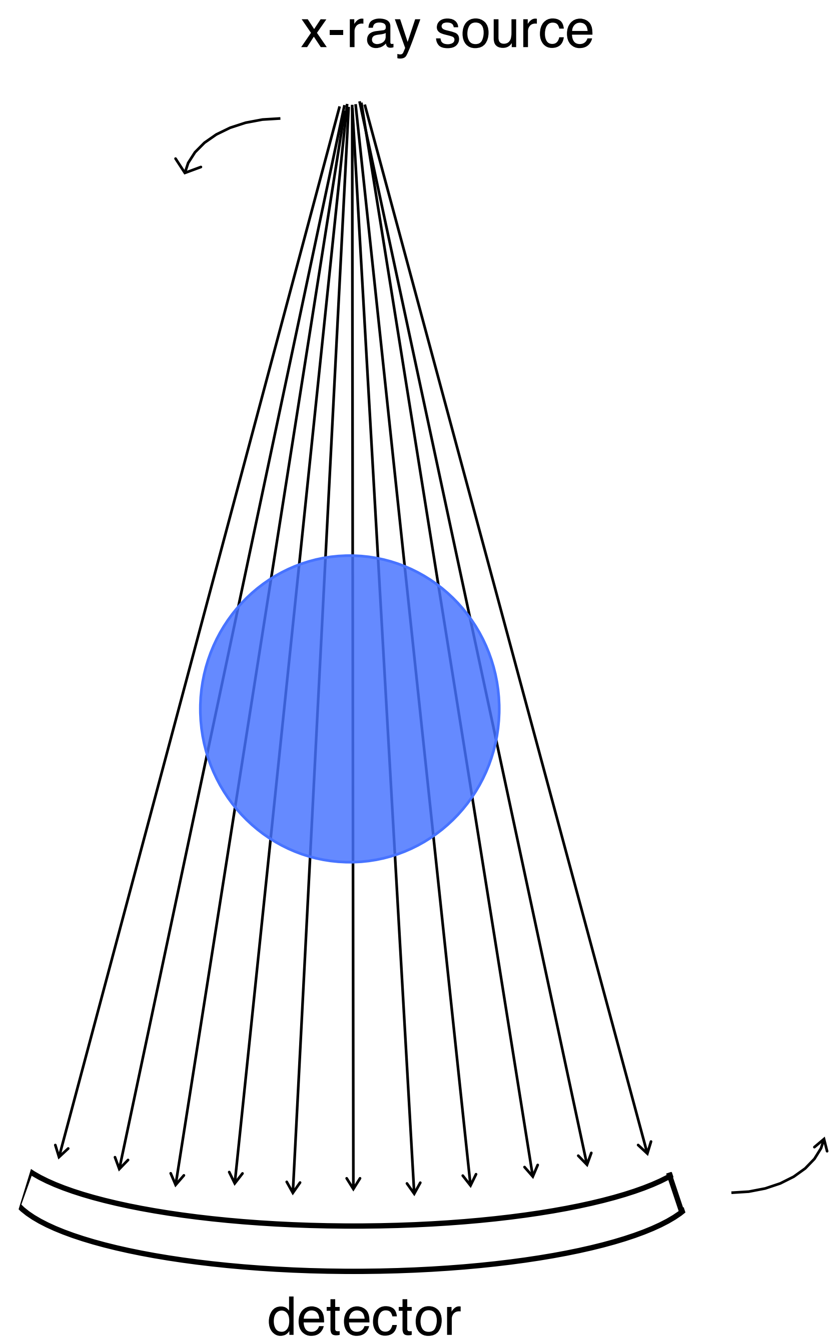


Top view

CT systems



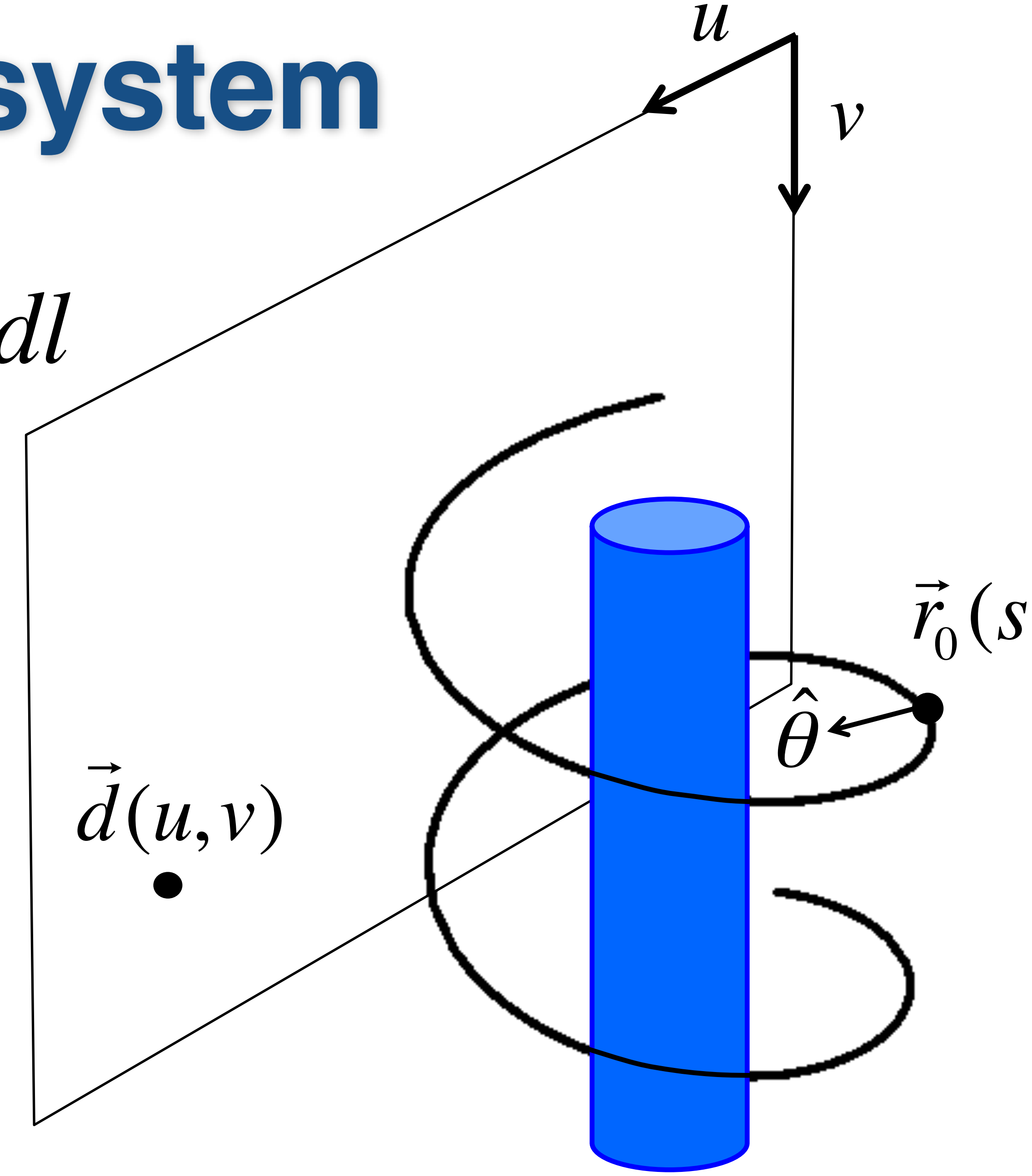
parallel beam



fan beam

Modern CT system

$$p(s,u,v)=\int_0^{\infty}\mu(\vec{r}_0(s)+l\hat{\theta}(s,u,v))\,dl$$

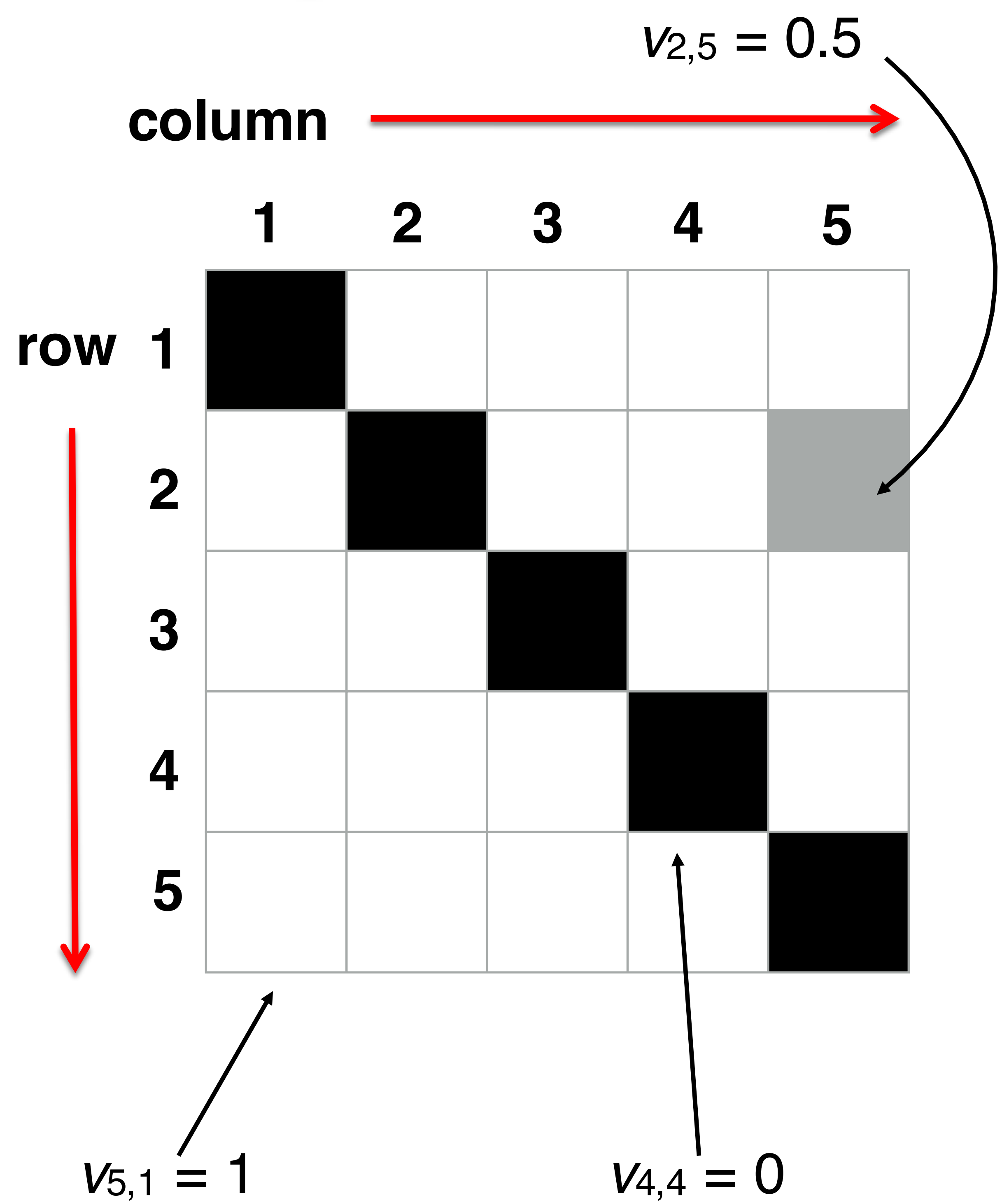


Helical configuration

Break

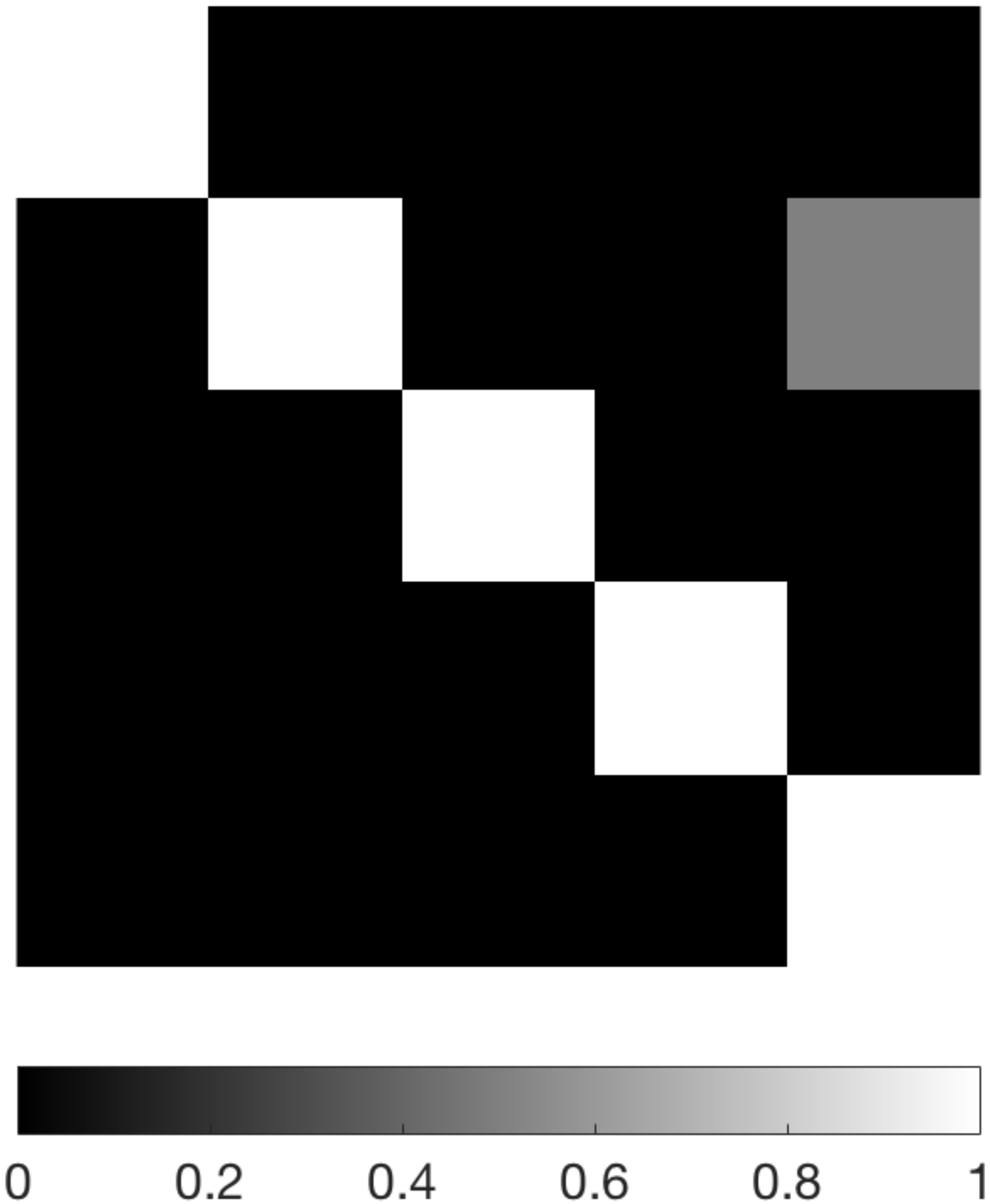
What is an image?

- An image is a matrix
- A pixel $v_{i,j}$ is a value in a matrix
- For a simple greyscale image, black=0, white=1, and numbers in-between represent greys



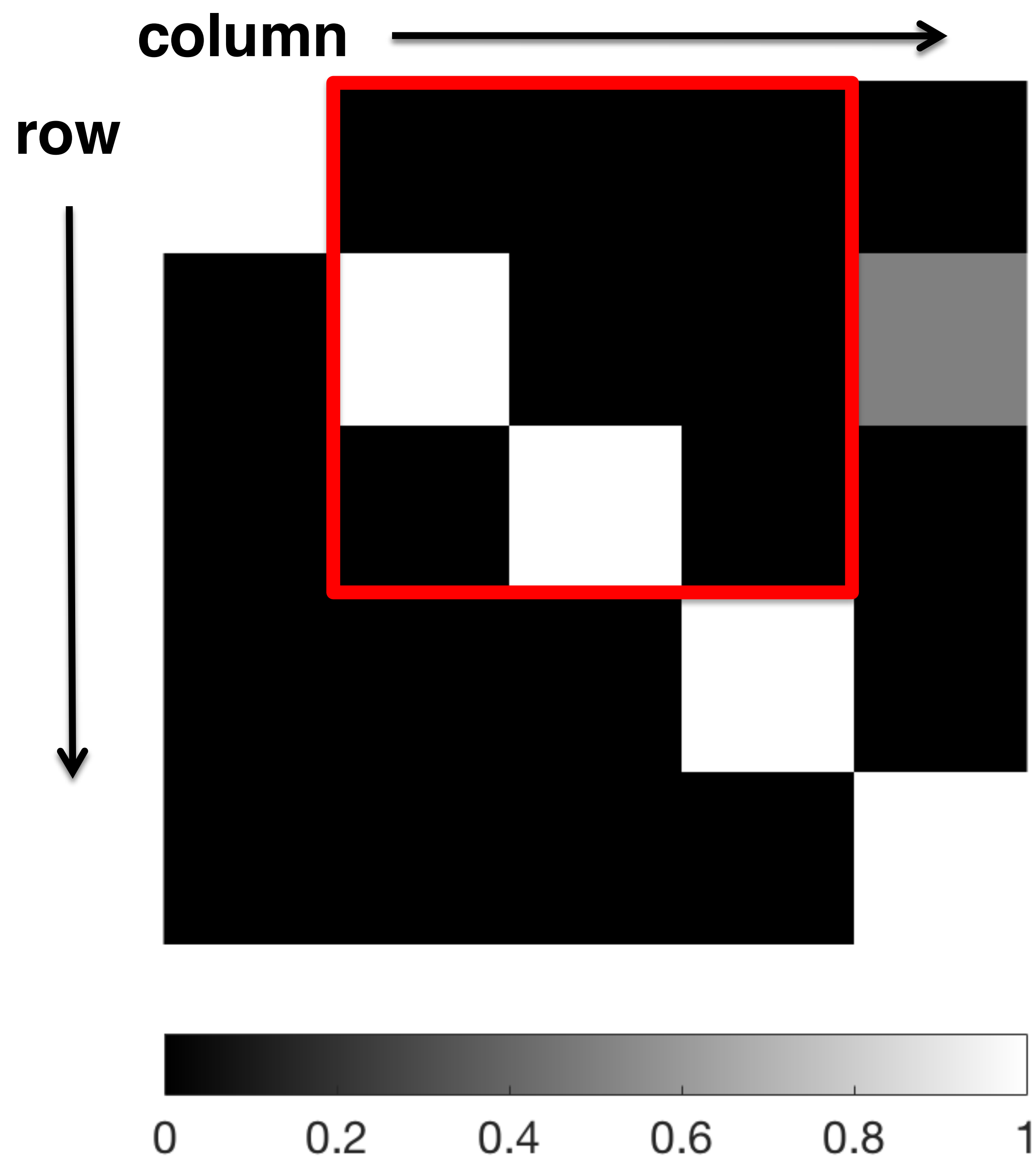
Matlab: simple image

```
clear all;
close all;

N = 5;

V = eye(N);

%identity matrix:
%1 = white, 0 = black
imshow(V, 'InitialMagnification', 'fit');
colorbar

%change element (2,5) to gray
V(2,5) = .5;

imshow(V, 'InitialMagnification', 'fit');
colorbar('southoutside', 'Fontsize', 15);
```



Matlab: simple image

```
>> V(1:3, 2:4)

ans =

     0     0     0
     1     0     0
     0     1     0
```



Real images

http://www.telegraph.co.uk/news/newstopics/howaboutthat/11963318/Uber-delivers-cute-kittens-to-stressed-workers-on-National-Cat-Day.html

Kitten in Matlab

cat in Matlab

```
clear all;
close all;

I = imread('cat.jpg');

Idouble = im2double(I);

imshow(Idouble);
title('cat in Matlab')
```

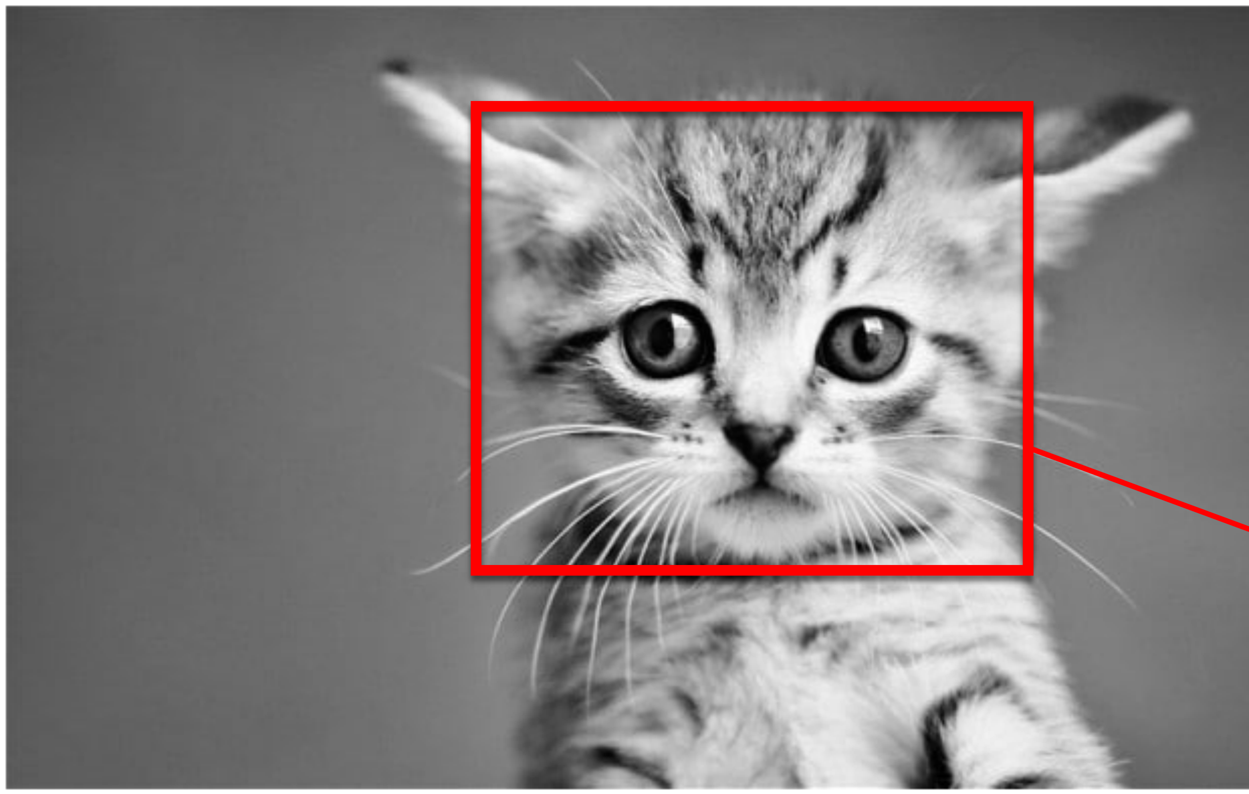
How big is the image? Type whos.

```
>> whos
  Name            Size                    Bytes  Class

  I             387x620x3                719820  uint8
  Idouble       387x620x3               5758560  double
```

Kitten is a 387 by 620 by 3 matrix (the third dimension is for color)

Kitten in Matlab

black and white cat



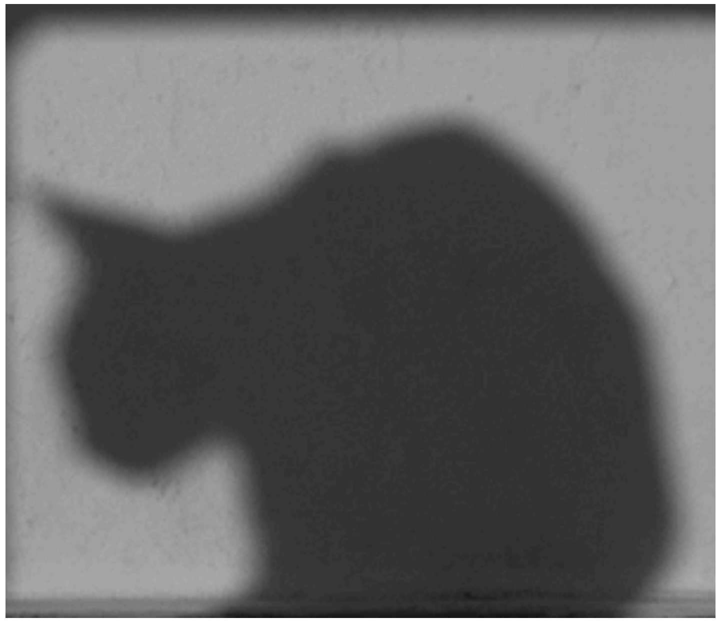
```
Ibw = rgb2gray(Idouble);

imshow(Ibw)
title('black and white cat')
```

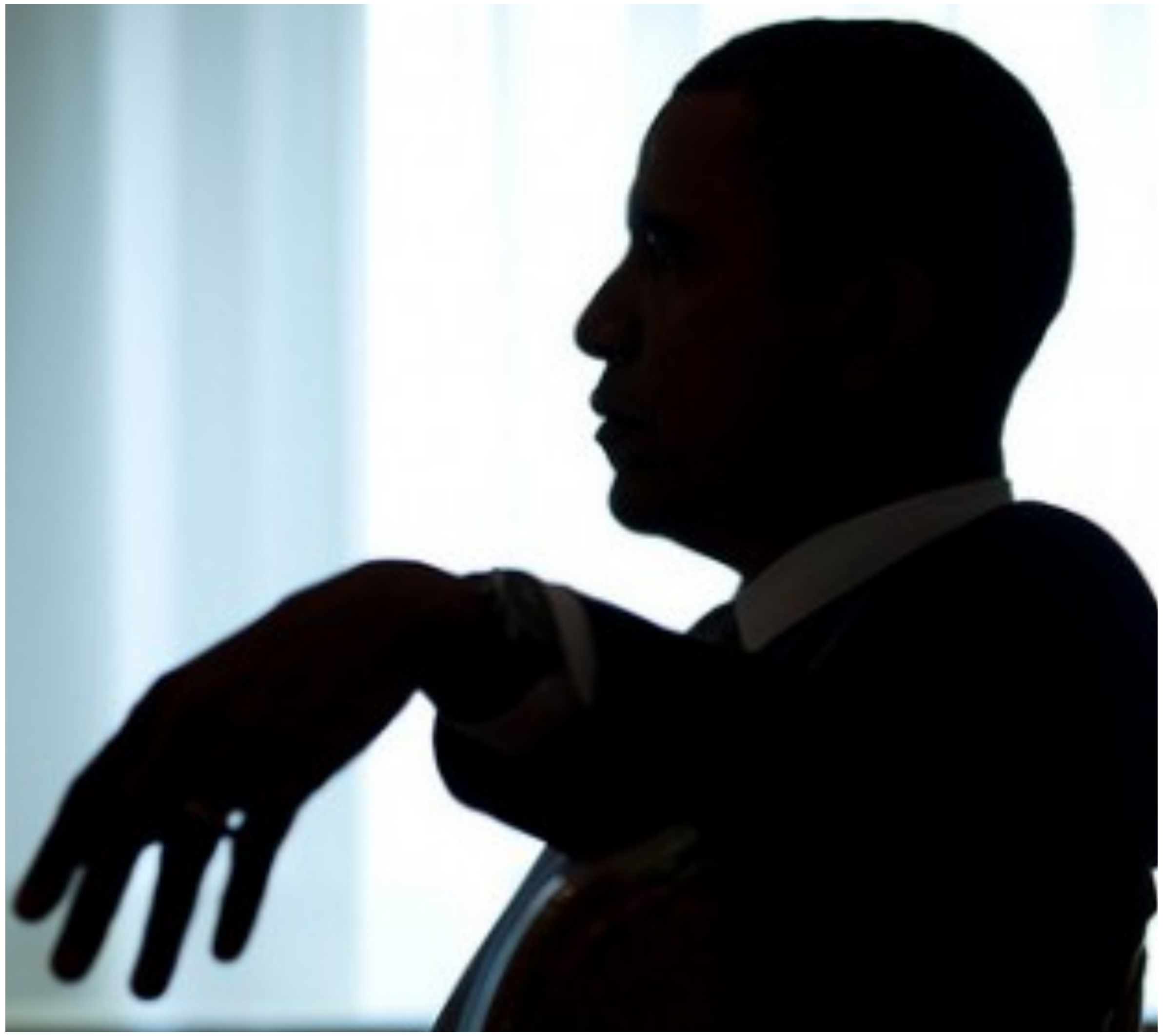
```
Icrop = Ibw(60:300, 200:500);
imshow(Icrop)
```

Name	Size	Bytes	Class
I	387x620x3	719820	uint8
Ibw	387x620	1919520	double
Idouble	387x620x3	5758560	double

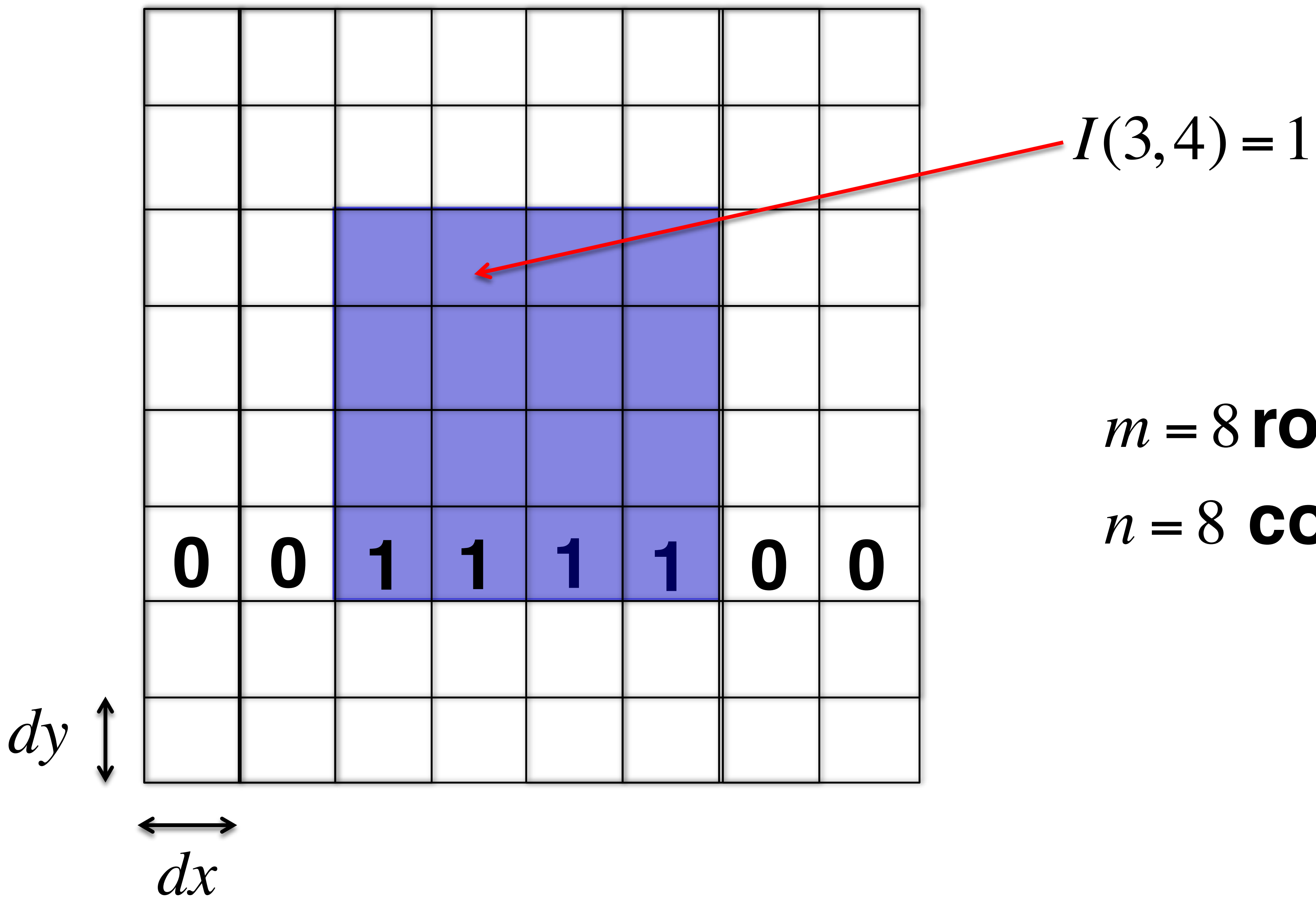
Black and white kitten is 387 x 620

Partial information: shadows





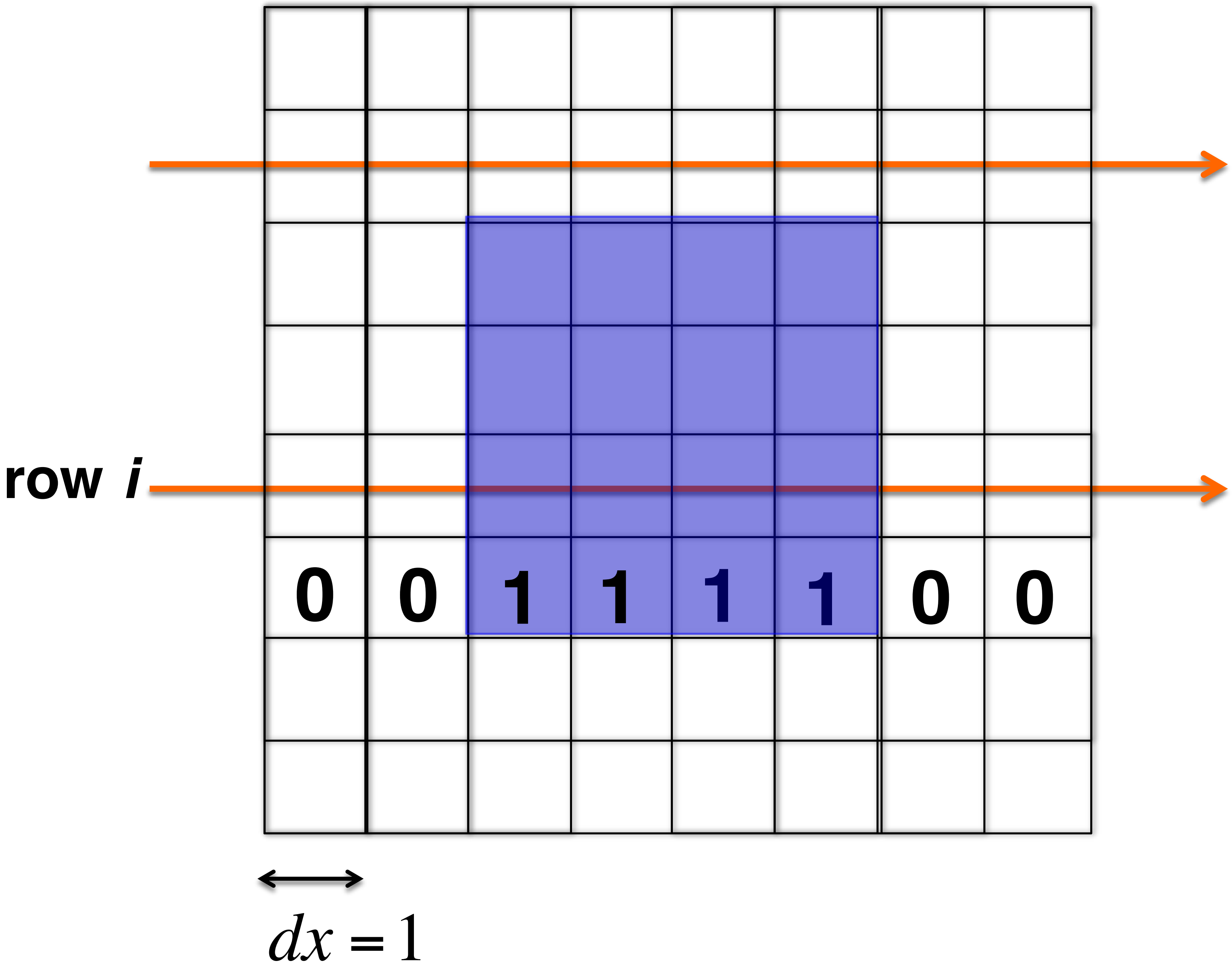
Radon transform



$m = 8$ **rows**

$n = 8$ **columns**

Radon transform



$$\sum_{j=1}^{n} I(i,j)dx$$

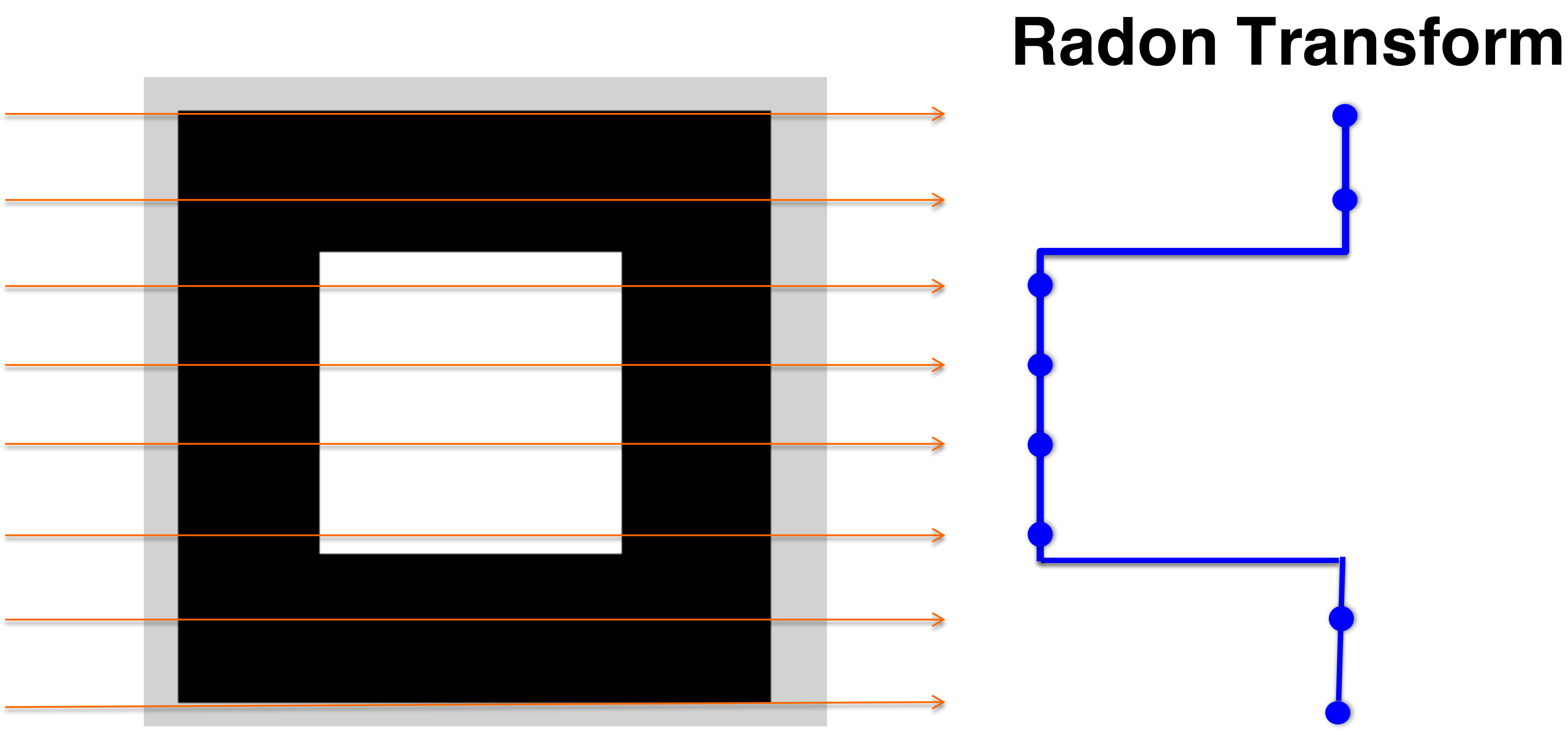
$$= I(i,1)dx + \ldots + I(i,8)dx$$

$$= [I(i,3) + \ldots + I(i,6)]dx$$

$$= 4dx$$

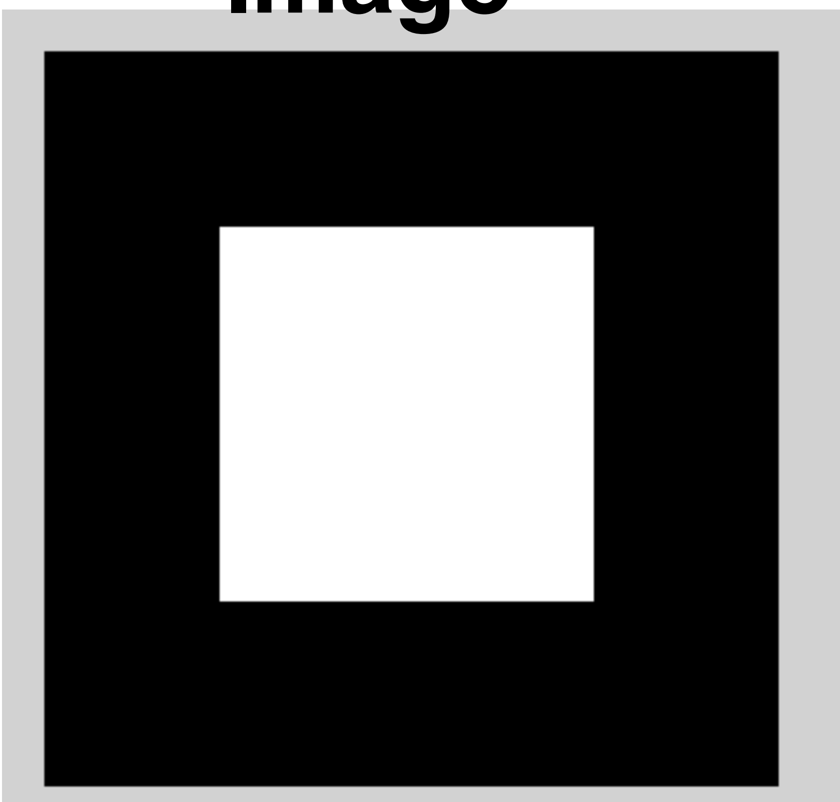
$$= 4$$

Shadows: Radon transform



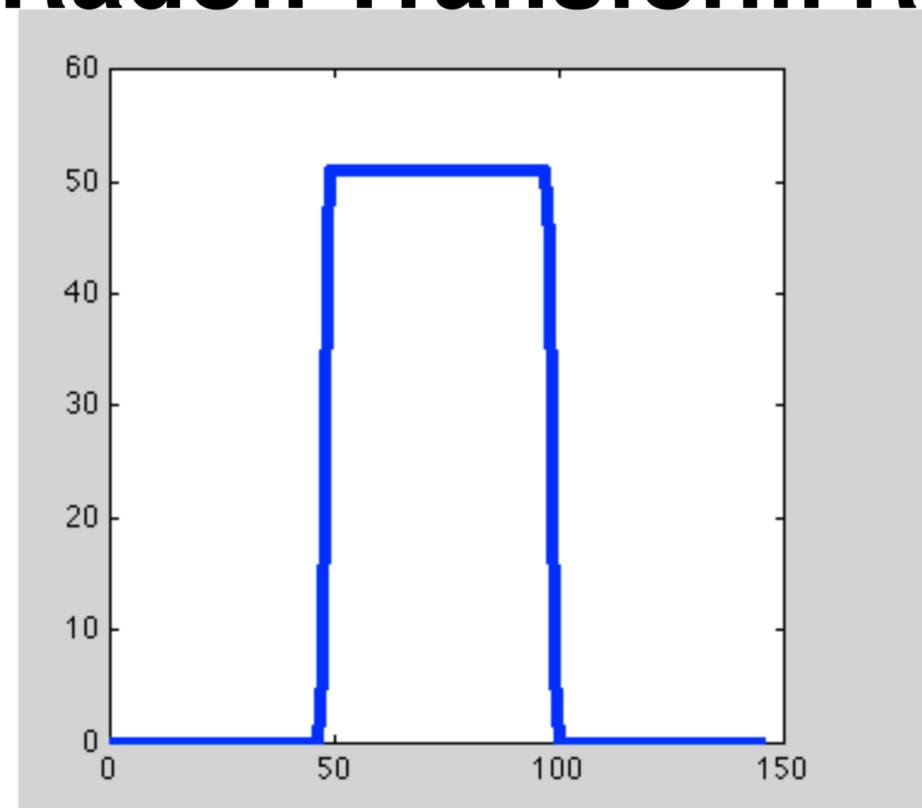
Radon transform

Image

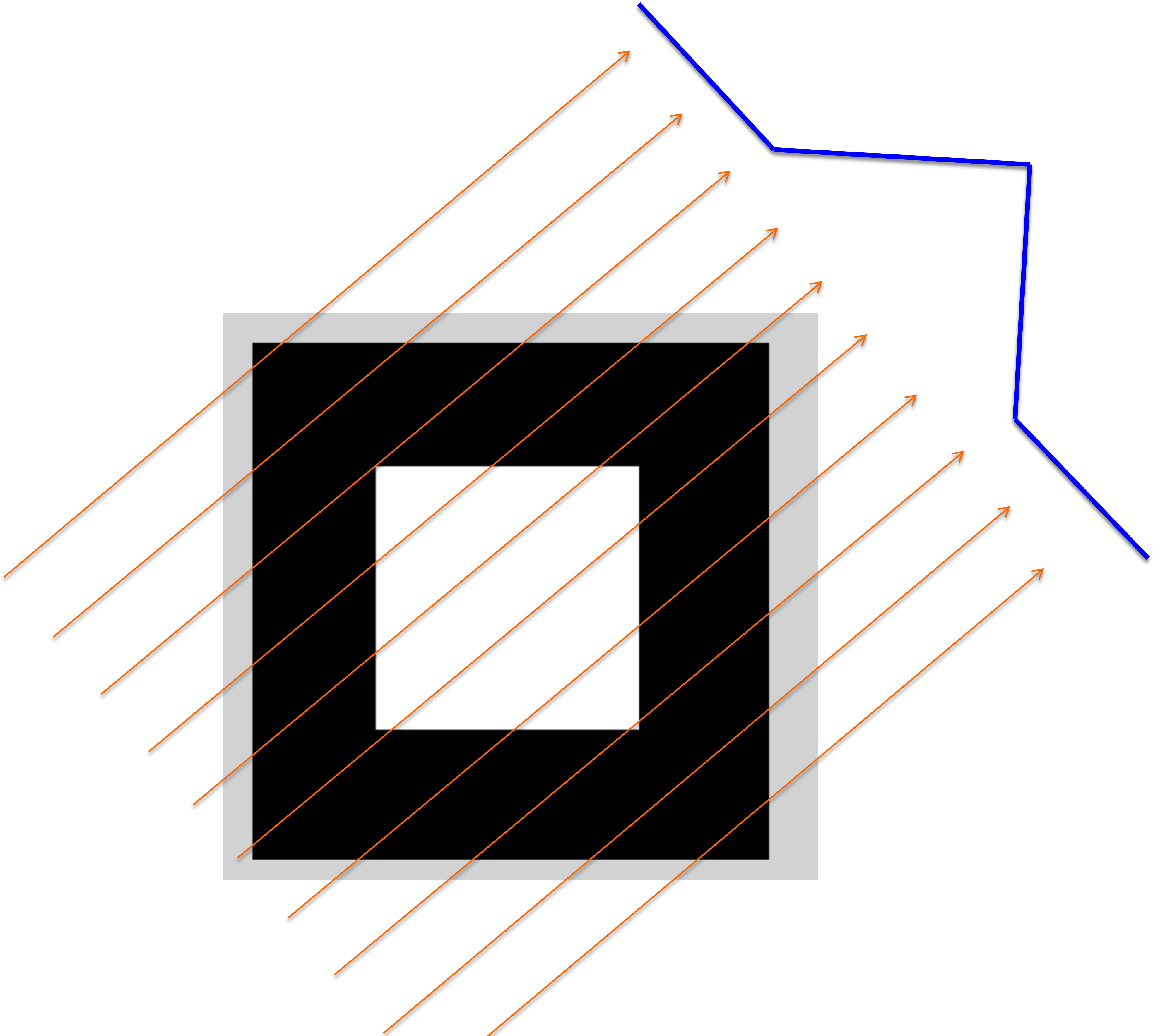


```
I = zeros(100,100);
I(25:75, 25:75) = 1;

figure(1)
imshow(I)

R = radon(I,[0 45]);
figure(2)
plot(R(:,1))
```

Radon Transform R

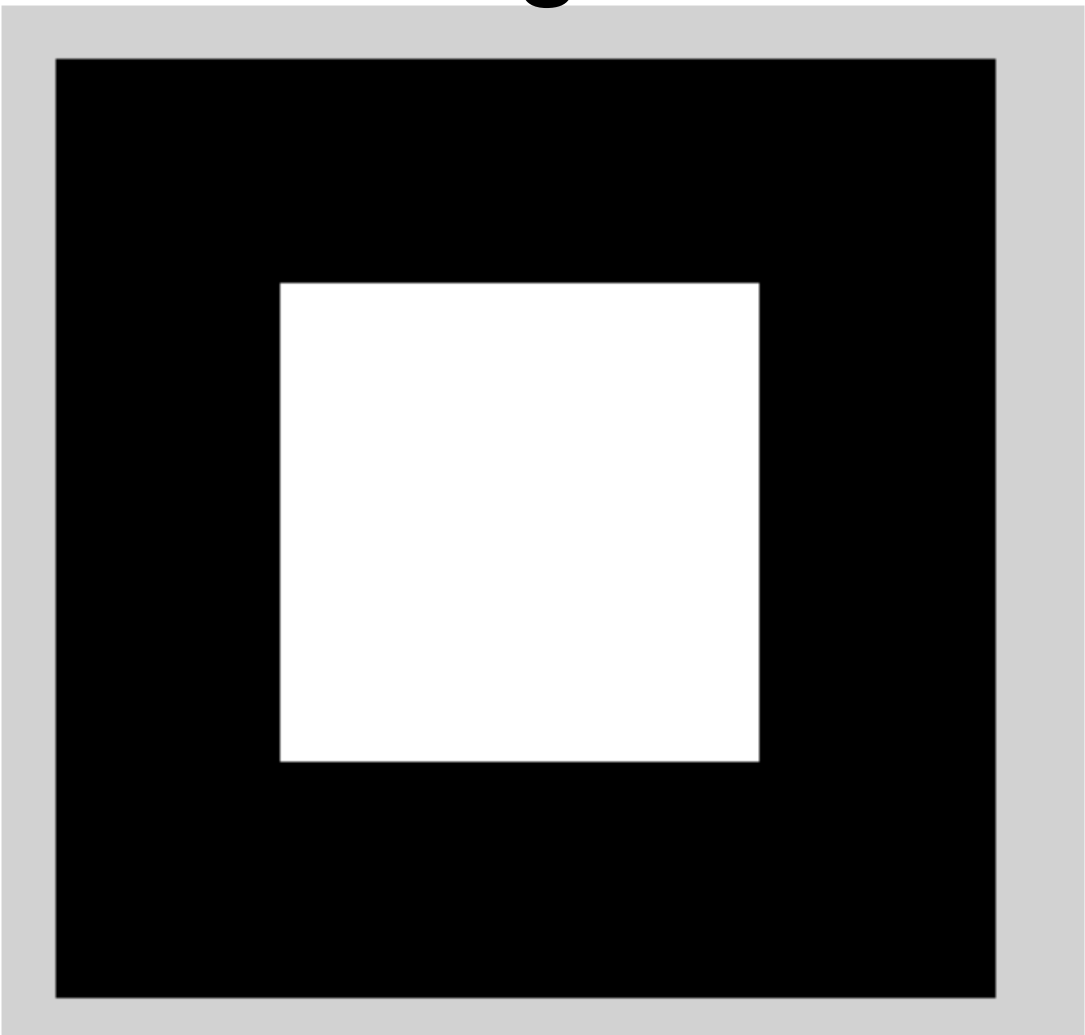


Second view: Radon transform

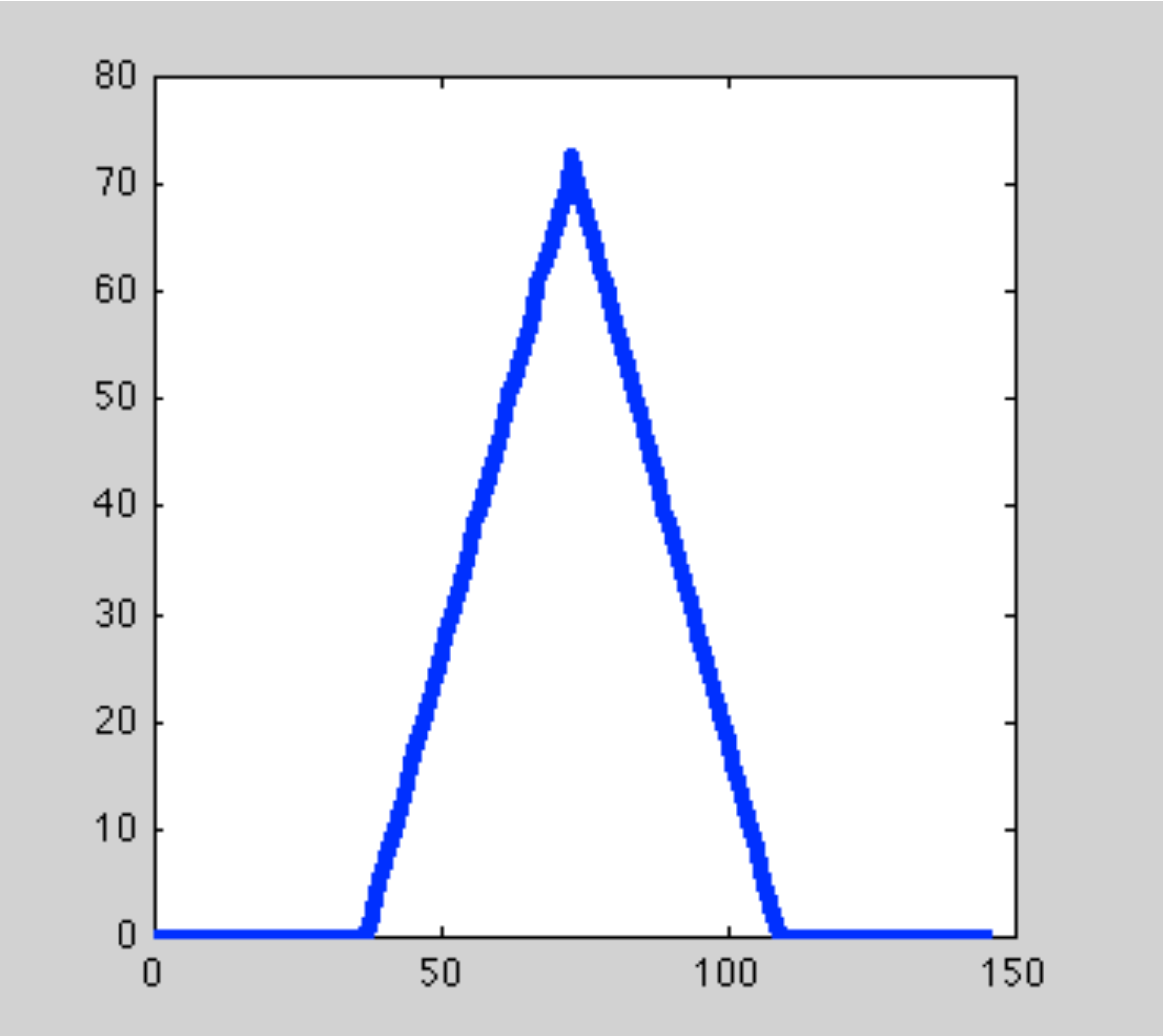


Second view

Image



Radon Transform R




```
figure(3)
plot(R(:,2))
```

Lab next Tuesday

- Images and Radon transform in Matlab
- Bring your computer!